

7.26 Lec 1 -

☆ fill out recitation form

Introduction to Host-Pathogen Interxns

- ① History
- ② The Cause of Disease?
- ③ Koch's Postulates + Bradford-Hill Criteria

- testing for diarrheal disease
 - sequencing
 - cultures
- "evidence of variables"
 - how did we come to understand germs?

3 theories of Infectious Disease

① Humoral Theory

- 4th c. BC - 1850
- Hippocrates in Ancient Greece
- 4 humours:
 - blood, phlegm, black bile, yellow bile
 - "balance" of humors = health
- Food directly creates humors
 - mostly treating symptoms
- Bloodletting, Leeches
- Regular Enemas, Vomit-Inducing Machines
- 1670's was the discovery of the bacteria!
 - extremely long-lived theory

② Miasma Theory

- 4th c. BC - 1880
- "bad air" / pollution
 - belief that disease is from the environment
- moving away, perfume to cover bad smells, close windows
- don't believe in patient-to-patient transmission
- sewage (dump in the river)
 - fecal-oral contamination

The Rise of Germ Theory

- 1854: John Snow

- cholera ← contaminated water
- map to trace disease
- epidemiological approach
- accepted in 1866

- 1850s: Florence Nightingale

- Founder of Modern Nursing
- worked on battlefield in Crimean war
- implement practices to prevent 2° death
 - handwashing, ventilation, sewer clearing, pest + unsanitary clothing removal

- Joseph Lister

- Listeria + Listerine
- Antiseptic surgical techniques
 - carboxylic acid
- suggest microbes cause disease
 - NOT real evidence

- exams aren't cumulative
- office hours by appointment

3 exams:

- viruses
- bacteria
- eukaryotes

- Pasteur

- fermentation caused by microbes
- heat kills microbes

- Robert Koch (1877)

- took the first photo of a bacteria (*Bacillus Anthracis*)
 - from correlation to causation
- Koch's postulates
 - required criteria

Koch's Postulates

- ① suspected Agent in all diseased organisms and not in healthy ones
 - e.g. anthrax in diseased mice only
 - ② agent must be isolated from organism and grown in pure culture
 - e.g. petri dish
 - ③ agent must cause disease in 2nd healthy animal
 - ④ same agent must be re-isolated from 2nd animal
- limitations?

- some organisms only cause disease in certain individuals
- viruses can't be grown in pure culture
- obligate intracellular pathogens

Bradford-Hill Criteria

- for inferring causality

e.g. Zika Virus (2015)

- ① Strength of Association
 - strong correlation of Zika Infection vs. Microencephaly
- ② Consistency
 - reproducibility across different cities
- ③ specificity
 - unusual characteristics of Zika vs. Rubella
- ④ Temporality
 - harder w/ Zika b/c birth defects take so long to manifest
- ⑤ Dosage/Biological Gradient
 - No gradient in Zika (binary infection or not)
- ⑥ Plausibility
 - e.g. found Zika Viruses in brains
 - Zika can kill brain progenitor cells
- ⑦ Coherence
 - does epidemiology match w/ lab experiments?
- ⑧ Experimental Criteria
 - clinical intervention: mosquito nets for Zika
 - nets are very wide-spectrum
- ⑨ Analogy
 - are there analogous infections to provide basis?

7.26 Lec 2 - Viral Structure + Entry

- what problems do viruses need to solve

1. Essential Characters
2. Viral Structure
3. Entry into Host Cell

- why care about viruses?

- ① Infect all living things
- ② Extremely Abundant
- ③ Viral Fossils = 40% of the human genome

- co-opt viral functions

· huge driver of morbidity/mortality

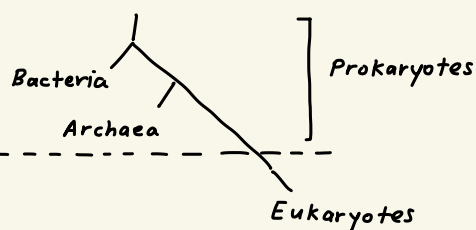
- what is a virus + how do they cause infection?

- A) Essential characteristics of viruses
- B) Viral Structure
- C) Entry
- D) Replication
- E) Exit
- F) Host Response

"molecular battle b/w host + virus"

A) Essential characteristic of Viruses

- nucleotide-based, requires other machinery



- no common ancestor for viruses

- evolved independently times
- HUGELY diverse

- exceeds host diversity

- genome can be any nucleotide structure
- fast speed of evolution

- what is a virus?

1) Behavior:

- can only reproduce within a host cell

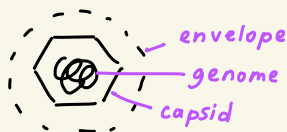
- 2 essential processes

- genome replication (host or viral machinery)
- viral particle protein translation (host machinery)
- no virus has its own ribosomes

must interact intimately w/ host

2) Parts:

- genome
- capsid (protective shell)
- envelope proteins (optionally)
 - lipid membrane stolen from host



- viral scale:

- @ least 1/10 size of organelles
- some are the size of large proteins

- what are the central problems the viral structure has to solve?

1. Protect genome (esp. environmental insults)

2. Disassemble upon entry to correct cells

- strong but not too strong

3. Fully contain the genome

- size of genome vs. complexity of capsid
- capsid must be as simple as possible

- How does the virus use its structure to solve the 3 problems?

3. capsid needs to pack genes

- HSV 150Kb genome - 125 nm capsid
- mitochondria 16 kb - 200-700 nm

① Direct contact b/w genome + protein capsid

- capsid can have $\oplus$ charged (K, R, H) aa's and attract $\ominus$ charged nucleotides
- have to neutralize charge to get self-repelling genome to pack

② special nucleotide-binding core proteins

- both condense + decondense DNA
 - attached to capsid + genome $\rightarrow$ splays out genome when envelope explodes thereby ensuring timely transcription

③ Packaging by host cellular proteins

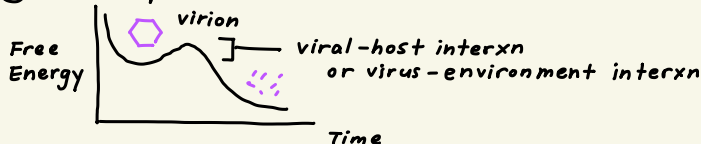
- borrow eukaryotic histones
- only possible in polyomavirus (dsDNA) genome
- helps evade host immunity b/c genome looks native
- virus doesn't have to encode genome packing proteins

1. Strong enough to protect the genome

① repetitive structure

- rod/helical or icosahedral
- have to be made by very few repeating subunits
- very structurally sound

② virus capsids are metastable



2. capsid must dissociate appropriately

① required for viral entry

- ensure replication @ the right time

- Viral Entry

1. Random collision b/w virus + host cell

- likelihood of collision determines required infection load
- determines how a virus is transmitted
- overall capsid strength determines infectiousness

2. specific binding b/w protein + host receptor $\rightarrow$ tighter interxn

- protein/protein or protein/glycan $\rightarrow$ determines host cellular tropism/susceptibility
 - tropism/susceptibility = types of cells that can get infected
- major therapeutic target
 - can prevent viral entry

- how do you figure out host receptor of virus?

- start w/ host cells the virus can infect:

- do a very large CRISPR experiment

viral particles + $\rightarrow$ cell death (susceptible)

$\downarrow$ genome wide KO of surface proteins

cell library $\leftarrow$ sequence

dead dead alive dead $\leftarrow$ sequence

- sequences that are enriched against baseline are needed for infection
- start w/ cells that the virus definitely can't infect
- overexpression screen

○ cells that cannot be infected

↓
library of overexpressors

↓ sequence

↓ add virus

↓ sequence

↓ genes that are missing are the correct genes

- not too burdensome (only one organism)

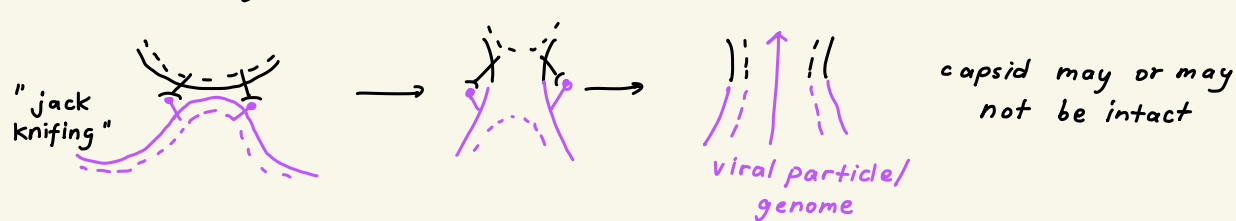
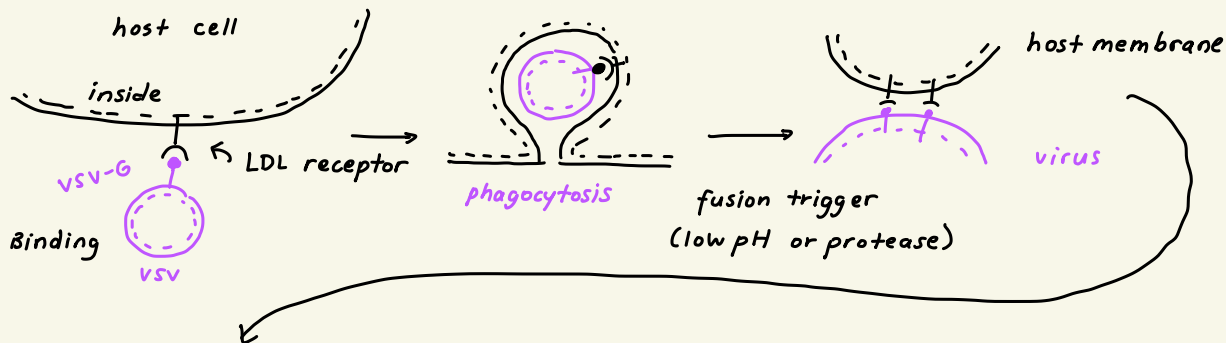
- test cross reactivity by expressing human receptors!

- viral entry (cont.)

· enveloped viruses

ex. vesicular dermatitis

host cell



- function of fusion (fusogen) is separate from binding domain

☆ · Key to pseudotyping viruses

- viruses move within the cell

· cannot diffuse through the cell

ex adenovirus escapes the endosome to get to the nuclear pore

· sometimes capsid proteins have to bind to cytoskeletal components

- uncoating / disassembly

· often triggered by low pH in endosome

ex. semliki forest virus - capsid binds directly to host ribosome

- positive ssRNA (exact same substrate as ribosome!)

ex. adenovirus uncoats when binding nuclear pore

next class: viral genome replication!

- virus pseudotyping

· changing viral surface protein to change viral tropism

· works best if virus is enveloped

· maintain inner machinery but change innate tropism

- why is pseudotyping useful?

① gene therapy - deliver DNA to any host cell

② research - decrease danger of virus being studied

· prevent accidental infection

③ basic science - search for protein-protein interxn

→ successful interxn = virus enters the cell
(determine entry by sequencing approach)

7.26 Lec 3 - Viral Genome + Replication

- Review Session at 9 AM FRIDAY FEB 28

- Host Requirements

- Viral Genomes (+ implications)

- Principles of viral Genomes

- coding strategies of viral genomes

1. What are 2 essential functions of viral genome?

① code mRNA for all essential viral proteins

- must get converted to mRNA to be compatible w/ ribosomes

- how does the virus coopt replicatⁿ machinery?

- how do many kinds of genomes get converted to mRNA?

② template for more genomes

- how does virus coordinate timing b/w mRNA prod + genome prod.?

2. What are the 2 essential host requirements for viral replication?

① susceptible

- host cell has to have the right surface receptor to mediate viral entry

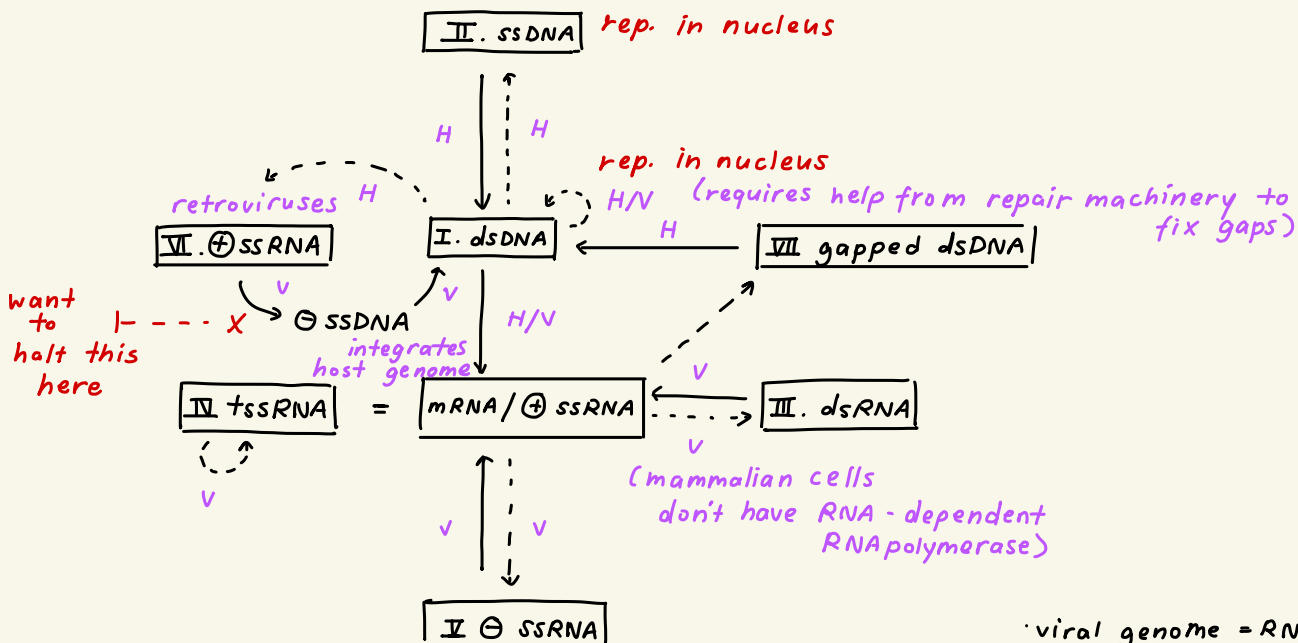
② permissive

- host cell is able to support viral replication

- if you can neutralize either of these, you can prevent infection

8. Viral Genome Classification

- all viral genomes must become mRNA/⊕ ssRNA



- RNA to RNA conversion only happens w/ viral enzymes

- more +RNA viruses than any other genome type

① RNA polymerase does not have proofreading func. → RNA viruses are "quasi-species" (powerful evolutionary forces)

- $10^4 \times$ higher error rate than DNA polymerase

② High diversity of RNA-guided RNA polymerases

- they just have to make more genome copies for the virus, antivirals are HARD to develop

③ virus needs ways to distinguish its own RNA from host RNA

- how does it know what to package?

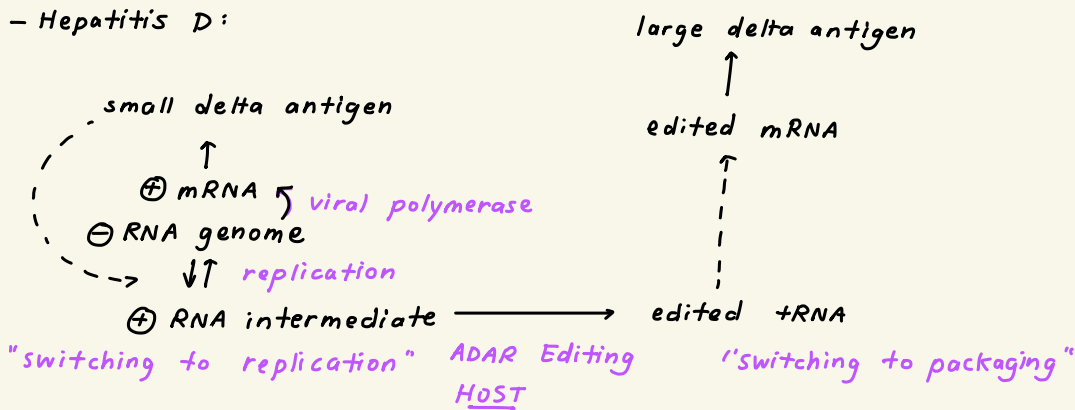
• extra features might also be recognized by a host anti-viral response

- In bacteria, most viruses are DNA viruses. Why?

- nucleus of eukaryotes created an additional barrier for DNA viruses
- DNA viruses are more stable

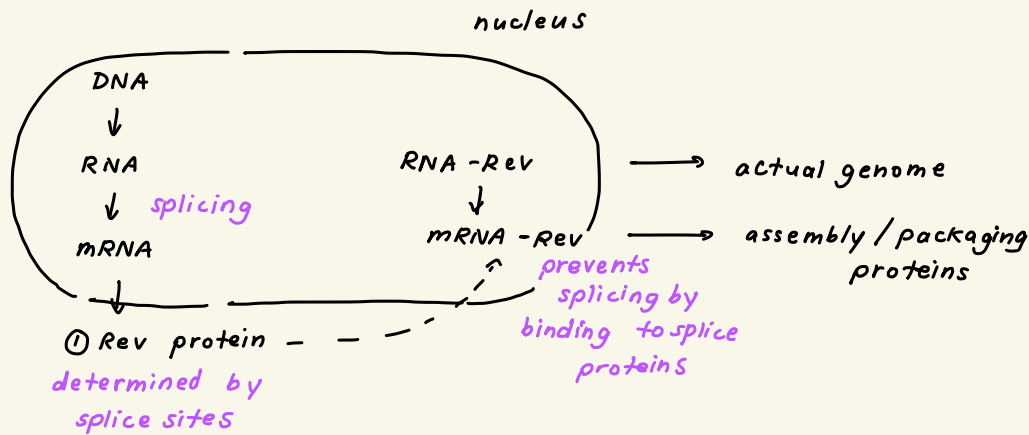
D. Coding strategies of Viral Genomes

- what do viral genomes encode or not?
 - DON'T: ribosomes, energy prod., membrane synthesis
 - DO:
 1. signals for timing of gene expression + replication
 2. parts for viral proteins
 3. needs to modulate host defenses
- small genomes require clever coding strategies

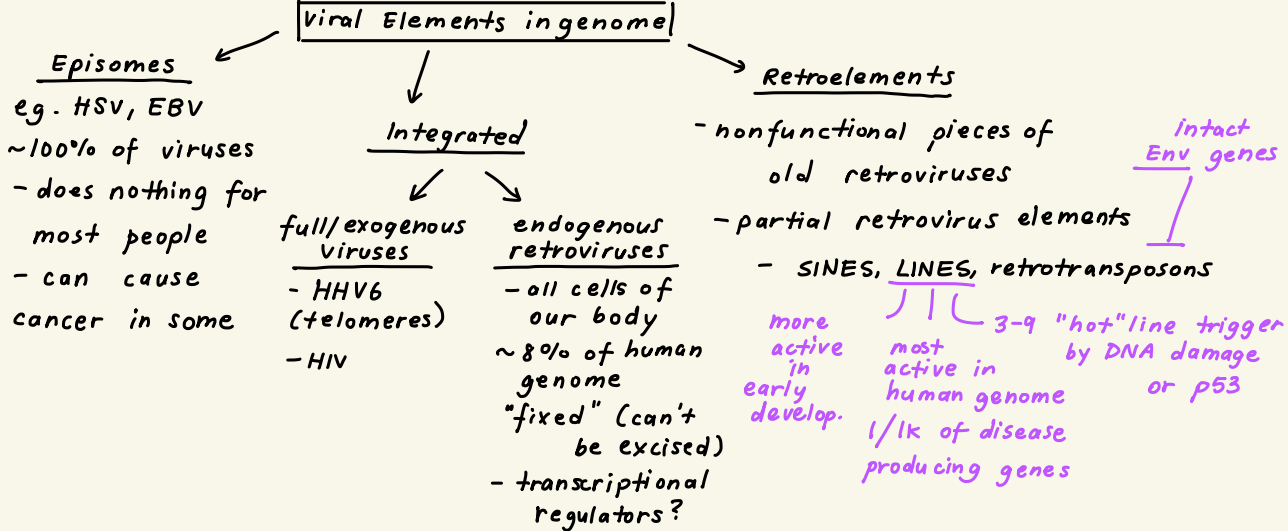


- ① protein is for replication
- ② once enough copies exists, the virus swaps to packaging
 - layers multiple proteins into same space

- Alternative splicing in HIV:



E. Endogenous Proviruses + Retroelements



- Implications of viral fossils in human health?
 - Retroelements are critical to evolution
 - cause of 1/2 of the inversions since speciations
 - syncytin-1 $\leftarrow$ envelope gene
 - have fusogens
 - mediates placental formation by controlling fusion of epithelial cells
 - mediate fusion + development of osteoclasts
 - formed by repeated fusion of precursor cells
 - commensal bacteria activate ERV expression
 - activates innate immune pathways
 - generates an "immune tone"
 - major cause of inter-individual genetic variation
 - Retroelements cause disease
 - 15% of CF mutations
 - excess ERV = neurodegenerative diseases
 - high reactivation of ERVs during sars-cov-2
 - highly correlated to severity/ cytokine storms
 - How are these things regulated?

7.26 Lec 4 - Viral Pathogenesis + Host Response

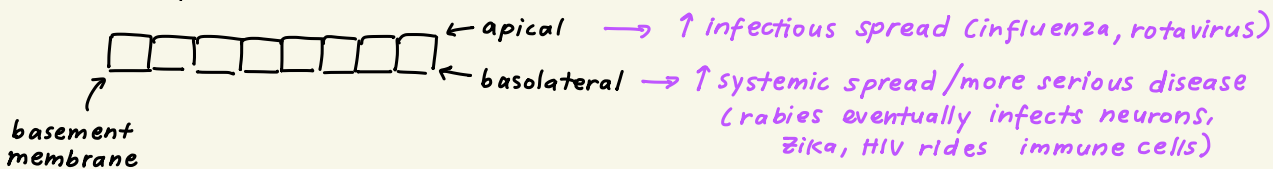
- A. Requirements for a viral infection
- B. What happens inside a viral-infected cell?
- C. Mechanisms of Host Defense

A. Requirements for a Viral Infection

- cells need to be susceptible + permissive
- but what about organisms?
 - ① sufficient # of viral particles reach the host
 - ② within host need to find cells that are susceptible, permissive
 - ③ host defenses are insufficient
 - otherwise all viruses would all be killed

B. What happens in virus-infected cell?

- ① Virus genome replication
- ② protein synthesis
 - virus suppresses mRNA expression in HeLa cells
 - why do viruses globally suppress host transcription/translation?
 - highly conserved behavior
 - necessary to get viral RNA to be able to compete w/ host
 - avoid defense host protein production
 - strategies to bias translation for viral proteins?
 - ① using alternate initiation factors
 - IRES (Internal Ribosome Binding Site)
 - mimics tRNA-mRNA structure
 - mRNA subsequence
 - ② cap-snatching
 - viruses have proteins that yank 5' 7-methyl-guanosine from host mRNA and puts it on viral mRNA
 - ③ localization of viral mRNA w/ translation machinery
 - statistically more likely
- ③ Rewiring host cell physiology and metabolism
 - ex. Adenovirus E1A rewires 30+ cellular signaling proteins
 - also rewires glucose metabolism
- ④ Viral Escape
 - Direction of viral release influences disease phenotypes
 - ex. epithelial cell



- in healthy blood donors, ~65% of the time, people see HHV7
- 1% CMV
- 72% EBV

released basolaterally

C. Mechanisms of Host Defense

① Physical Barriers

- Epithelia
- peristalsis/cilia/secretions (send outside things back out)

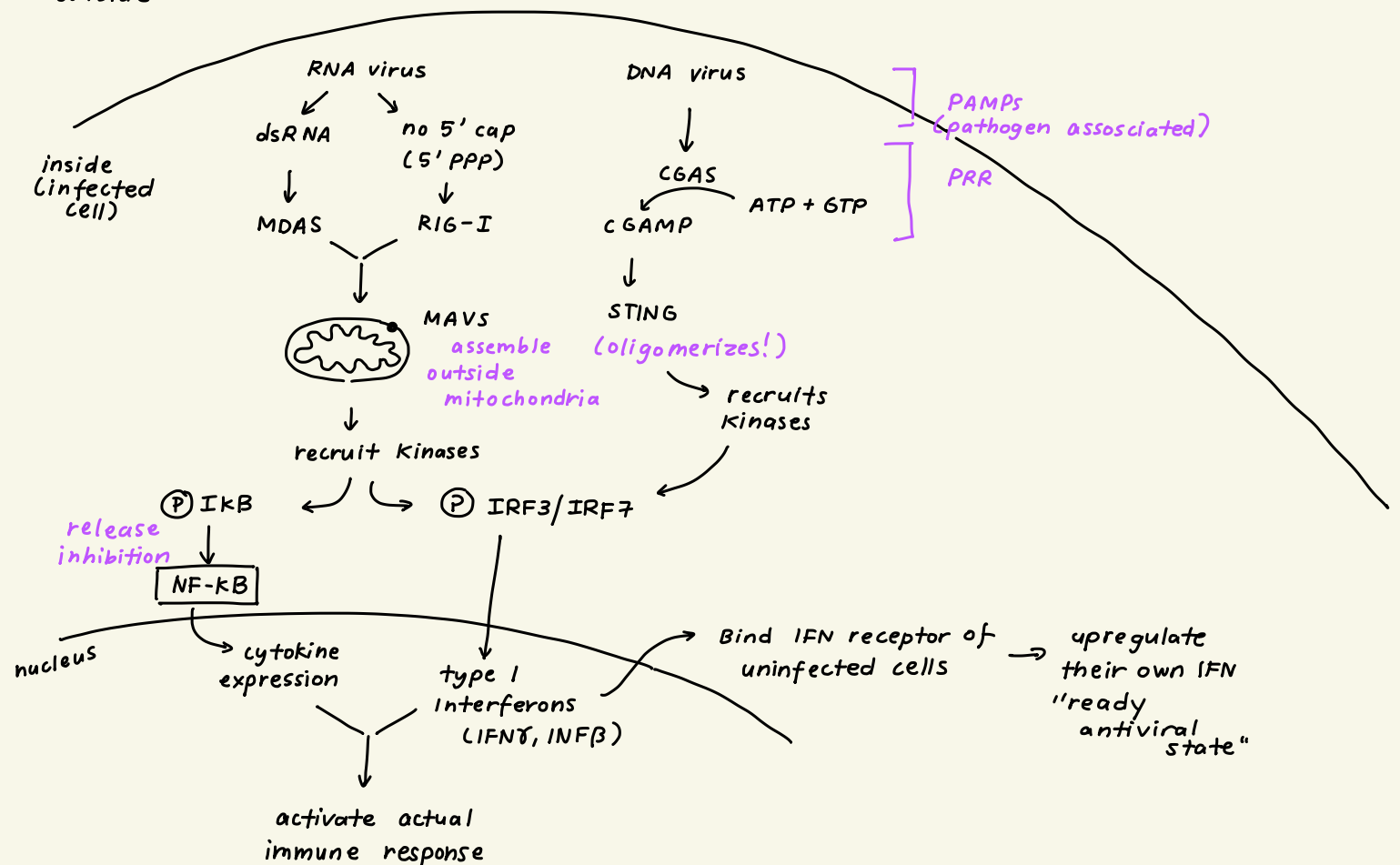
Flu → neuraminidase → clears acid in mucus (surface)

✗
Tamiflu

- ② Innate Immune Defense :
- 3 components
 - ① pattern recognition receptors
 - present in all cells
 - first things to recognize viral parts
 - recognize:
 - DNA in the cytosol → note that location is important
 - dsRNA
 - ② response in cell
 - PRR binds ligand
 - ↓
 - signal transduction cascade
 - ↓
 - turn on TF or caspase
 - ↳ apoptosis
 - ③ signalling between cells
 - ① soluble factors:
 - cytokines → "program activation"
 - chemokines → "cell migration"
 - small proteins that change behavior of receiving cells
 - ex. interleukins are cytokines
 - ② cell-cell contacts
 - innate immune cell ↔ target cell
 - CNK ↔ infected cell
 - innate ... ↔ adaptive ...

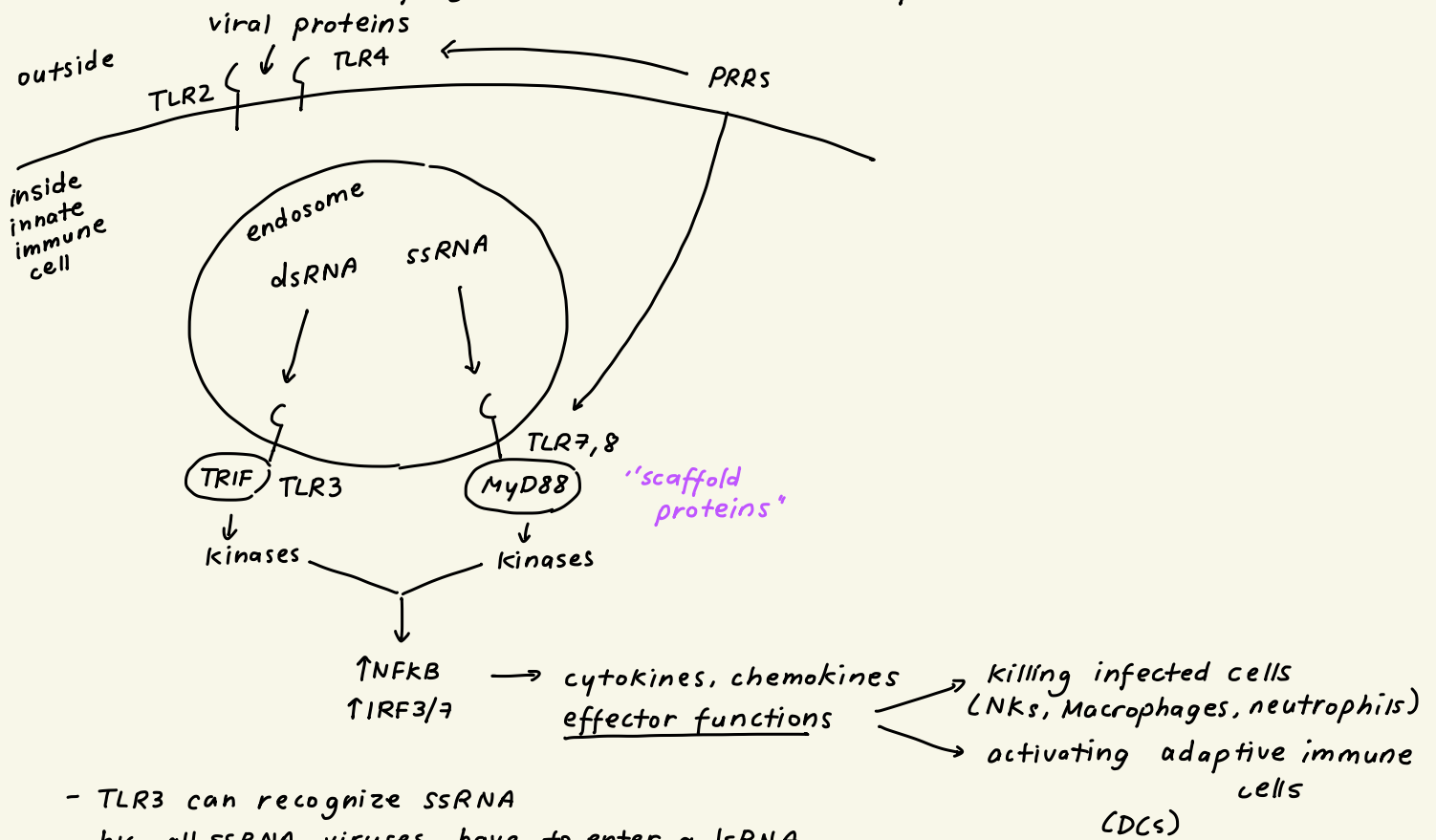
ex.

outside



ex. innate defense in professional immune cells

- dendritic cells, macrophages, natural killer cells, neutrophils



- TLR3 can recognize ssRNA

b/c all ssRNA viruses have to enter a dsRNA state during replication

· Can recognize all RNA viruses

next lecture = adaptive immune system

7.26 Lec 6 - HIV-specific Cytotoxic T cells

HIV

- CD4200 = AIDS Biomarker
 - ↑ viral Load = ↑ Infection Speed
 - ↓ CD4 = ↑ Infection Severity

Challenges to Effective

- integrate to chromosome
 - invisible to immune system + drugs while inactive
- heavily glycosylated HIV Envelope
 - need lots of SHM to neutralize
- extensive mutability + progressive immune escape

Mechanism of HIV control:

- SNP on chromosome 6 (immunity)
(MHC-I)
 - HLA presents viral fragment
- HIV controller GWAS
 - specific amino acids lining HLA peptide binding groove
 - also associated w/ autoimmunity!
 - Better cross-reactivity!

GWAS + HIV control?

- HLA + antigen → activate cytotoxic T cells
 - granulins kill host
 - prior to viral packaging

Epitope Presentation

- HLA Class I : 289 Residues
- side chains affect 3D protein structure

Social Network Theory: w/ HIV structure

- "structure-based network analysis"
- target CTL to highly networked epitopes

CFSE Assay:

CFSE-stained PBMCs → HLA-optimal HIV peptides → 6-day incubatⁿ flow cytometry

- HIV controllers respond extra to highly networked epitopes

- only intact provirus can reinitiate infection
 - intact HIV higher in HIV controllers
 - how do they control then?
 - clonally expanded in centromeric region, Z-finger
- elite controllers CTLs target highly networked epitopes
→ clearance of HIV proviral reservoir

Effective CTL eliminate infectious pro-virus

Cellular activation in controllers ⇒ no transcription

7.26 Lec 7:

Adaptive Immune System Trade-off

① Speed vs. Specificity

- innate immunity is the fastest
 - cell intrinsic response
 - professional immune cell response
 - give Adaptive Immunity more time
- Adaptive Immunity is happening simultaneously
 - Require multiple, proliferation + migration to tissue of interest

② Self Tolerance vs. Pathogen Detection

- BCR + TCR undergo random mutation & recomb to get specificity
 - ex. anglerfish males are absorbed by females
 - female anglerfish have no BCR/TCR recomb (no adaptive immunity)
 - high risk for autoimmunity

③ Memory Vs. Bias

- Memory B + T cells respond faster b/c they don't need priming
 - (no need for cytokines/costimulation)
- This creates bias within the immunological repertoire
 - ☆ · "original antigenic sin"
 - lose the advantage of memory in the first place
 - need a certain level of similarity
 - also relevant for vaccines

Therapeutic Conditions:

① Antigens neutralize the pathogen directly

- mark for destruction by professional immune cells
- direct antibody therapy works correctly
 - ex. 3 ebola antibodies given to outbreak patients
 - 5/7 patients survived BUT very labor intensive
 - ex. convalescent plasma for COVID didn't work
 - b/c they just pooled plasma w/o antibody analysis
- need enough Ab to coat the viral particle

② Best Vaccines should engage B & T cell response +

- ideally the mucosal immune response
- ex. live attenuated polio virus for immunization

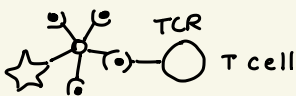
Important Experimental Considerations:

① Can easily directly count # of Abs against specific antigens

- BUT hard to link Abs to a specific B cell w/o sequencing

② Harder to count T cells specific to a particular antigen

- TCR doesn't just coat protein by themselves
- create peptide-MHC tetramers



TCR's are just lower affinity
TCR's function via avidity

- need multiple TCR's to have enough to form an immune synapse
- if this work, can do scseq to figure out what's happening

Viral Immune Evasion:

- A) Patterns of Infection
- B) acute, C) latent, D) persistent

How does the immune system generate specific responses against unknown pathogens?

- VDJ recombination (B/T cell repertoire)
 - double-edged sword = potential for autoimmunity
- have to still have self-tolerance
 - some viruses evolve to be similar to host proteins

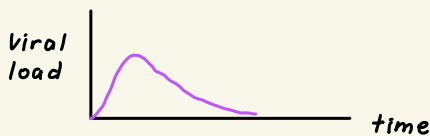
Weaknesses of Immune System that viruses can exploit?

- MHC are "eyes" of immune system
 - escape by hindering MHC peptide processing
- immune escape via repeated viral mutation
- interfere w/ PRR
- mimicking self-antigens

3 Patterns of Viral Infection:

① Acute

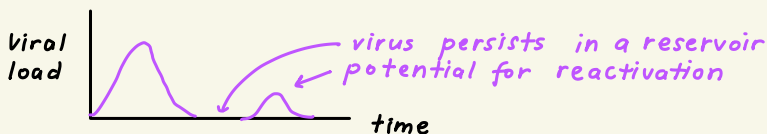
- defining feature:



- implication:
 - per person, the virus does not successfully evade the immune system
 - on a population level, the virus does successfully evade immune recognition
 - mutation of antigenic epitopes
 - exploiting delay in adaptive immune response

② Latent:

- defining feature:



ex. herpes simplex virus (HSV)
shingles/chickenpox

- implication for immune response:
 - productive immune response is the reason for latency
 - mechanism for evasion = mechanism for latent
- what kind of viruses can become latent?
 - dsRNA can't:
 - triggers acute inflammation b/c PRRs
 - ssDNA + dsDNA can
 - live in the nucleus to escape immune detection
 - only one kind of ssRNA can be latent
 - lack of capping in cytosol can make it detectable
 - retroviruses become latent by becoming DNA (integrated proviruses)
- how is a latent virus maintained in host cell?
 - ① nonreplicating host cell:
 - viral genome is nonreplicating

- HSV, VZV become stable episomes in neurons
- travel down + infect @ skin that neuron innervates

② replicating host cell

a) replicating episomes

eg. EBV in B cell

replication performed by host machinery

b) integrate into host genome

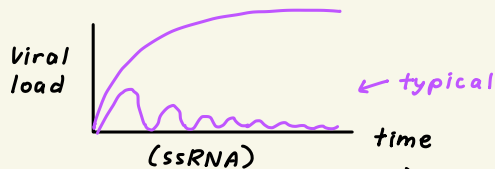
e.g. retroviruses

eg. HHV6 (rare non-retrovirus)

} source of viral fossils

③ Persistent Infection

- defining feature:



never reaches undetectable levels

← typical

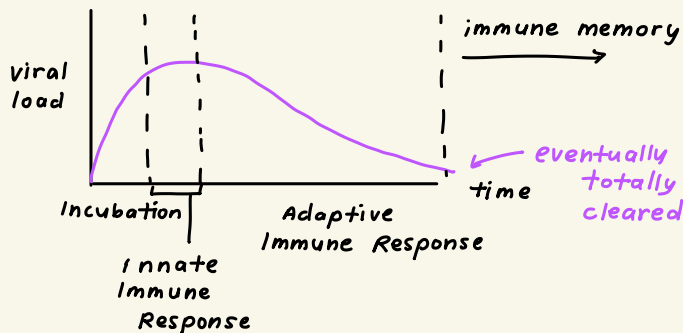
- hep B, hep C, HPV (dsDNA)
- (dsDNA)

- implication:

- constant immune evasion is required

B. Immune Evasion by Viruses w/ Acute Infection

- course of infection:



· incubation period: time b/w infection & symptoms

· infectious period: time of active viral shedding

- @ the population level, 2 major immune evasion strategy (antigenic shift + drift)

· Drift = slight alterations in antigenic proteins (due to mutation)

· Shift = major change in surface/antigenic proteins

b/c recombination

- segmented viral proteins are particularly good @ this

- reshuffle when an organism is infected by 2+ viruses

- more likely in high-density pops (industrial farming)

· DRIFT causes CONTINUED PRESENCE IN POP, SHIFT causes PANDEMICS

- antibody-dependent enhancement

· some immune memory but these Ab's are now non-neutralizing

· sometimes we get antibody dependent enhancement

- major cause of death (i.e. hemorrhagic fevers)

- actually facilitates infection of immune cells by causing uptake

- cause cytokine storms b/c immune cells recognize viruses + cause pro-inflammatory cytokines but virus is uninhibited

C. Immune Evasion in Viruses w/ Latent Infection

① Minimizing expression of viral proteins

e.g. >90% have latent EBV

- linear dsDNA virus, but maintained as a circular episome
 - harder to degrade circular genome
- during latency, all viral proteins BUT EBNA1 are suppressed
 - suppressed by noncoding RNA
 - EBNA1 attaches episome to host chromosome to piggyback during chromosome segregation
 - unclear how only EBNA1 is expressed

② Latency in Tissues w/ Low Immune Surveillance

- HSV, VZV latent in neurons
 - immune evasion
 - doesn't have to bother w/ replication
- Treatment strategy: "lock-in" latency
 - want complete suppression

③ Block processing + antigen presentation by MHC I

- can't completely block MHC I expression
- NK cells have an inhibitory receptor that binds to MHC I

④ ↓ MHC I, ↑ HLA-E

- HLA-E is an MHC I that presents signal sequences of other MHC I's
- inhibits NK cells + CD8 T cells!

⑤ produce entire viral decoy for MHC I

- HCMV viruses prod. a gene that shares 25% a.a. identity w/ MHC I BUT looks structurally identical to MHC I & present self-peptide
- Bad @ presenting viral MHC I
- MHC mimic is presented on surface of infected cells

⑥ EBV EBNA1

- gly-ala repeat motif (almost 50% protein)
- blocks the proteasome!
- attach this motif to any protein to stop MHC I presentation

Lec 8 - Viral Immune Evasion

- if latent EBV doesn't cause immune effects, is there even a problem for the host?
 - reactivation is of course a risk
 - EBNA1 (latency-maintaining gene) is anti-apoptotic!
 - EBV is often latent in B cells
 - major risk factor for lymphoma → still must become proliferative by themselves
 - activated B cells can resist natural contraction after immune-mediated proliferation
 - many latent viruses have similar anti-apoptotic molecules

D. Immune Evasion in Persistent Infection

- require methods to inhibit host immune system

ex. chronic hep C

- 80% of patients develop a chronic/persistent infection
- chronic inflammation → 25% of cirrhosis, 20% of hepatocellular carcinoma

① Inhibit signaling by innate immune receptors:

- NS3/4A protease: degrades components of PRR signaling

② Mutation of T cell epitopes (evade adaptive immunity):

slows adaptive immune response

- HCV is an RNA virus

- high mutation rate b/c RNA
- takes weeks to develop initial T cell response
- "mutational escape" exploits this
- selects for sequences that physically bind worse to MHC & sequences that are recognized by fewer naïve T cell precursors
 - delete self-reactive T cell, are closer to self-antigens
 - the virus is not "smart", this is just enabled by high mutation rate + natural selection

③ T cell exhaustion/dysfunction

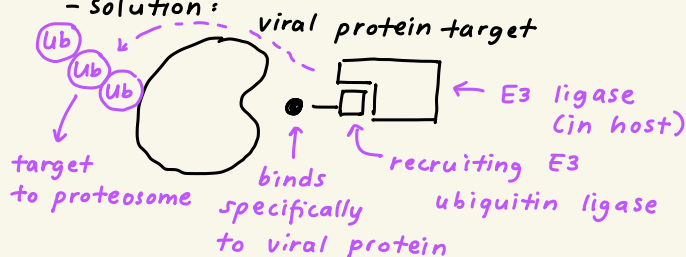
- T/B cells experience negative feedback via immune checkpoints
 - we have drugs to inhibit immune checkpoints for cancer treatment
- prolonged antigen exposure + TCR → increases expression of checkpoint receptor in T cell (PD-1)
 - ↑ signal coming through checkpoint → proliferative & cytokine/effector function in the T cell stops working
 - probably exists to prevent autoimmunity

Therapeutics & Viruses for Discovery

A. PROTAC-based antivirals

- PROTAC: proteolysis targeting chimeras
- problem: want to selectively degrade viral proteins
 - inhibit viral function
 - increase antigen presentation/rescue immune response

- solution:



- PROS: just need something that binds (doesn't have to suppress)
 - easier to design a binder than an inhibitor
 - enhances immune response
 - iterative (you already have the "warhead")

- CONS:

- poor membrane permeability/delivery
- 2 part molecule makes synthesis harder

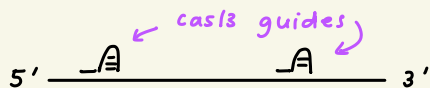
B. CRISPR Antivirals

- Problem:

- get rid of latent viral genomes
- change host factors needed by virus
 - change receptor required for host cell entry
 - gene therapy!
- specifically degrade viral RNAs (PAC-MAN)

- Solution:

- PAC-MAN: Prophylactic Antiviral CRISPR in human cells



- cas13 is an RNA-guided RNA nuclease

· PROS:

- easy to program specificity

· CONS:

- delivery of machinery is challenging

C. Lentivirus Engineering:

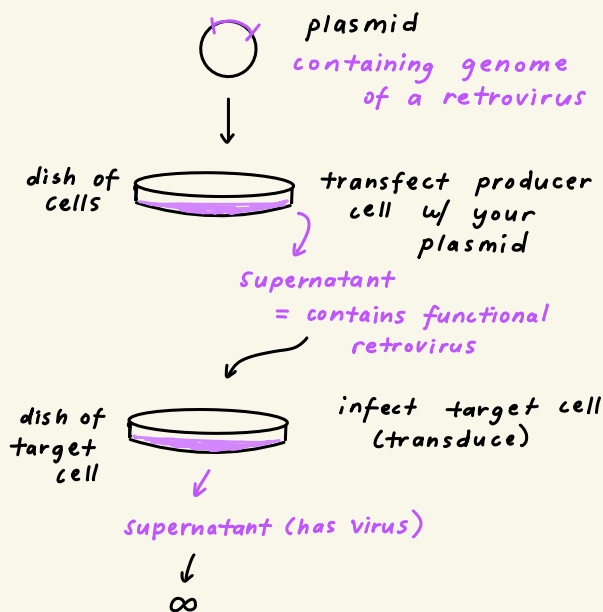
- retrovirus like HIV

- Problem:

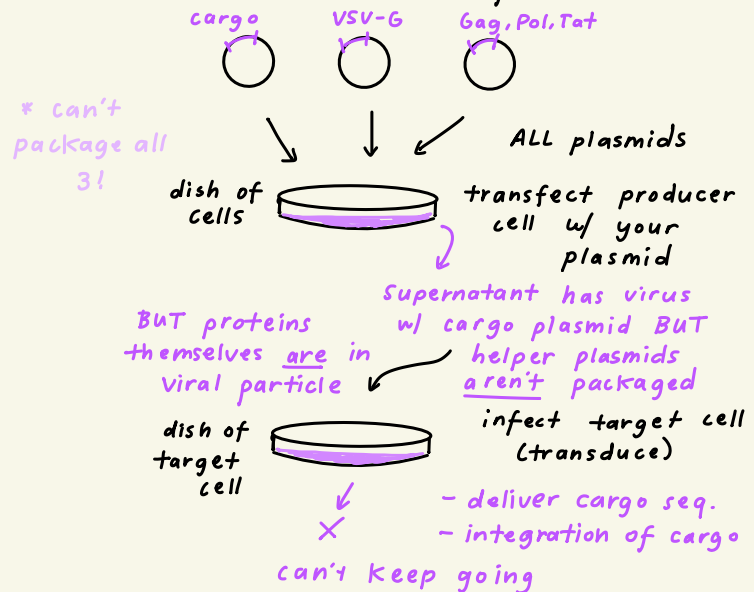
- integrating a specific sequence into the host genome

- Solution:

- modify retrovirus to deliver your sequence
- normal retrovirus:



· lentivirus: made it safer!



D. Adeno-associated viral engineering

- Problem: stably express gene w/o integration into host genome or siRNA (to turn down disease gene)

- much safer for clinical approaches

- Solution:

- AAV: ssDNA virus that turns into a stable episome
- cannot replicate w/o a helper virus
- broad tropism: can enter lots of host cells

- ~ 8 AAV-based therapies approved
 - retinal, spinal cord, liver, muscle
- target specificity is an area of active research
- might not be great @ repeated use b/c development of immunity
 - efforts to shield viral epitopes to make it more immune evasive

Lec 9 - Bacterial Pathogenesis

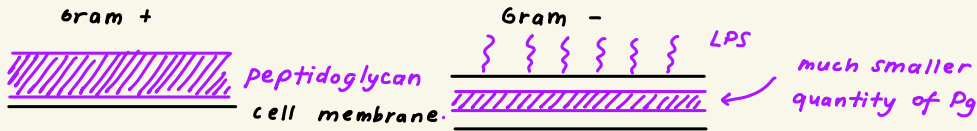
Key Concepts:

- 1) Bacterial & Niche Diversity
- 2) Transmission & Tropism
- 3) Becoming a Pathogen

Bacterial Feature:

① cell walls

- useful for visualization
- gram positive or gram negative?



- Gram stain
 - add crystal violet: bind peptidoglycan
 - add iodine: crosslink more in Gram +
 - alcohol wash mostly removes violet from gram - b/c little pg
 - saffranin as a counterstain (pink)
 - gram + is violet, gram - is pink

② Bacterial Shape (morphology)

- round = cocci (staph aureus)
 - grape clusters
- rod = bacilli
- comma = vibrio
- spiral = spirochete (aurelia)
- helical = helicobacter

③ sequencing/phylogeny

- 16S rRNA sequencing
- niche dependent phylogenetic microbe composition

commensals (normal flora) vs. bacteria

- pathogens disrupt normal host cell function, usually cause damage
- commensals have evolved to coexist with the host

how do commensals impact pathogen colonization?

- commensals & pathogens fight for space & nutrients
 - commensals can be a physical barrier to colonization
 - sometimes! commensals make nutrients that pathogens hijack
- what makes a bacteria successful?

- correctly finding a host, successful colonization

Tropism:

- generalists: broad specificity
 - ex. salmonella: chicken & people
- specialists: limited host range
 - ex. salmonella typhae specific for higher primates

Transmission/Access:

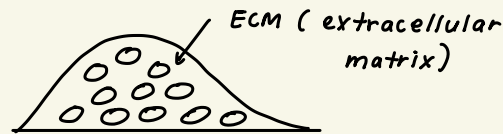
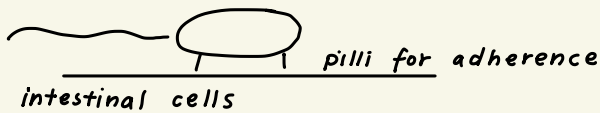
- Route-of-entry
 - fecal-oral · urogenital
 - respiratory · skin
 - ocular · blood & vector borne
 - defense: mucus, pH, tears
- vectors: mosquitos, ticks, leeches, needles, arthropods

Vibrio cholera

- aquatic pathogen
- transferred to human via contaminated water
- fairly fatal: bypass stomach acid to colonize the small intestine
- cause acute diarrhea:
 - rice water stool : die within hours
 - 1L/1 hr loss of liquid = 10^{10-12} vibrio
- wants to enter the water ASAP

Colonization: ex. Vibria Cholera in small intestine

- typical infectious load varies widely (dosage)
 - ex. 10 shigella can result in disease
 - ex. vibrio needs 10^4-10^6
- motility: need to migrate down the villi's mucus
 - ex. motile appendages = flagella
 - gram negatives + positives have similar engines
 - vibrio w/o comma shape have a hard time penetrating the mucus
 - swarming: swarm outward by $\uparrow$ flagellar length + #
 - prevent excessive crowding on intestinal surface
 - spirochetes have internal flagella
 - bacteria spirals as it moves
- adhesion



- creating a microcolony
- create a biofilm w/ 3D shape
- ex. massive biofilms in lungs of pneumonia patients

cell invasion

- ex. listeria & salmonella w/ intracellular life cycles
- limit competition w/ existing flora
- hide from some aspects of the immune response
- have very different

How do pathogens become pathogenic?

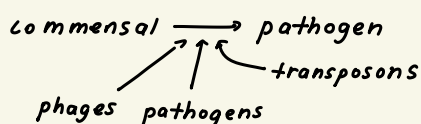
- commensals have become able to overcome a ton of barriers to become pathogens
- pathogens have smaller genomes than commensals
 - rely on host & get rid of unnecessary complexity
- experience both gene loss & gene gain!

gene loss:

- change selective pressure by existing in a host
- particularly true for specialists
 - need way fewer proteins
- most extreme example = obligates
 - quite metabolically dependent on host
- lose targets for immune system

gene gain:

- horizontal gene transfer: acquisition of mobile genetic elements



- acquired as large chunks called "pathogenicity islands"
 - genomic regions b/w 10 + 200 KB

- only in pathogens
- GC content is different from the rest of the genome

ex. *Vibrio* began as a non-pathogenic aquatic organism

① VPI (TCP) = toxin coregulate pilus

- helps the organism colonize & stick to intestinal surface
- also created a binding site for phage
- CPI (cholera toxin phage)

- integrates toxin production to cholera genome

1961 ② *Vibrio* 7th pandemic island

- in Haiti
- island helped increase fitness & epidemic spread
- want to both get + regulate the genes

- quorum sensing

- seen across many types of bacteria
- allow bacteria to communicate w/ one another
- help regulate group bacteria!

ex. *Vibrio fischeria* in bobtail squid

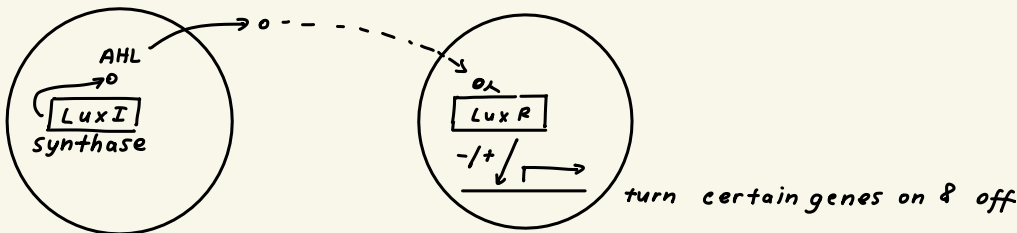
- species-specific chemical languages
- persistent infection:

- activating quorum sensing = $\uparrow$ Biofilm formation, $\uparrow$ swarming, $\uparrow$ virulence factors

- acute infection (e.g. *Vibrio*)

- $\oplus$ QS: $\uparrow$ dissemination genes, $\downarrow$ virulence factors
- such bacteria are focused on leaving

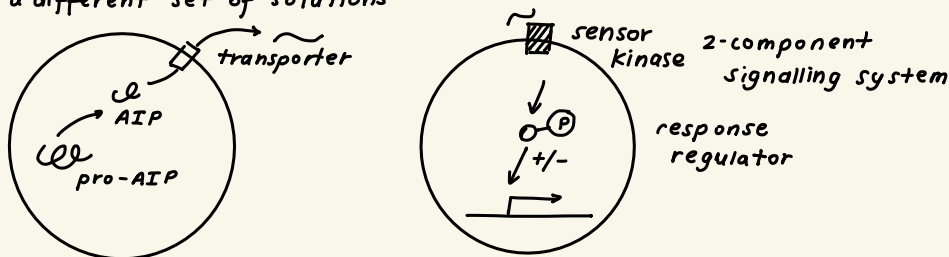
Gram $\ominus$



- very simple interaction circuit

Gram $\oplus$

- a different set of solutions



- very simple signal transduction
- these come in many varieties
- species-specific signalling molecules w/ repeated patterns!

7.26 Lec 11 - Bacterial Secretion Systems:

- ① single vs. double-membrane spanning systems
- ② substrates & delivery mechanisms
- ③ *Yersinia Pestis*

- allow things to cross membranes

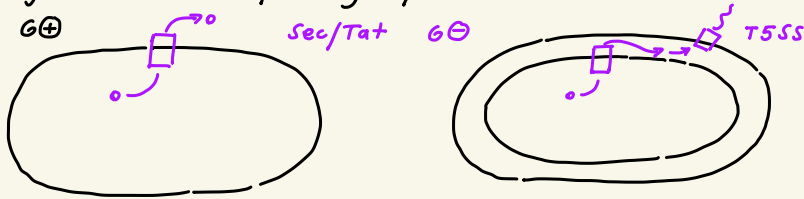
- variety of systems b/c bacteria use a variety of methods to interact w/ substrate
- do things to support movement of these proteins

- gram positive bacteria have 1 membrane

- require single membrane spanning secretion system

OVERVIEW:

Single-Membrane Spanning Systems:

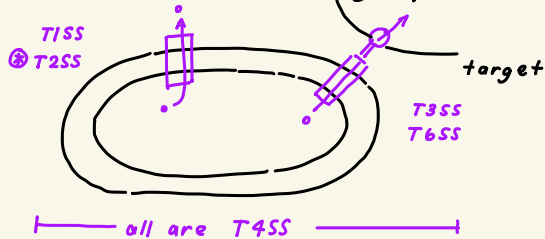


- systems for allowing cytoplasmic proteins to exit

- enter the periplasm in Gram ⁺

- gram negative bacteria proteins that need to exit periplasm can embed themselves into the outer membrane

Multimembrane-Spanning Systems:

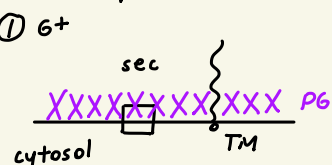


- T3SS can actually build a channel into another cell!

Sec/Tat Transport System:

- cytosolic proteins
- sec is highly conserved prokaryotes → animals
- Tat is conserved in prokaryotes → plants
- particular rules abt proteins:
 - hydrophobic sec 4EG signal sequence cleaved
 - co-translational & post-translational products threaded through
 - must be unfolded b/c pore is small
 - needs to be folded outside
- tat transports folded proteins only
 - twin arginine transporter
 - all substrates have SRR (twin arginine)

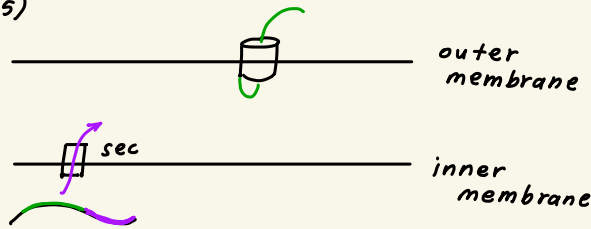
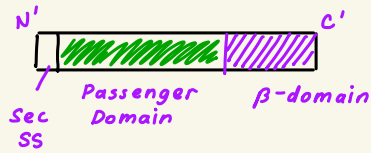
Sec-Dependent Surface Proteins - Adhesins, Enzymes...



- sortases covalently attach surface proteins
- LPXTG is a sorting motif

② classical autotransporter (TSS5)

- Gram $\ominus$



- passenger domain is the functional bit

- β portion folds in periplasm and catalyzes entry into outer membrane

T1SS

- simplest multimembrane transport system

- 3 parts (proteins)

- 1-step secretion: goes through inner & outer membranes in one channel
- only translocate unfolded proteins w/ C' terminal 66X6D signal



- pore-forming toxins don't oligomerize/assemble until later
- any substrates for pathogen & nonpathogen interxn

T2SS:

- 15+ proteins

- 2-step secretion

- ① synthesize toxin & use Sec or Tat pathway to deliver to periplasm
 - folds in periplasm
- ② piston-like delivery, ATP-dependent
 - pushes the toxin out
 - unclear how loading works here

Yersinia Pestis:

- historically important (Bubonic Plague)

- gram $\ominus$ bacteria

- requires multimembrane transport system

- vector transmitted

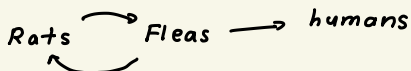
- fleas are the vectors of pestis

- 1347-1351: killed 25 million people

- 30-50% of the European population

- death in less than 1 week

- now, 1000-2000 cases a year
- often due to interxn w/ rodent kill, rodents



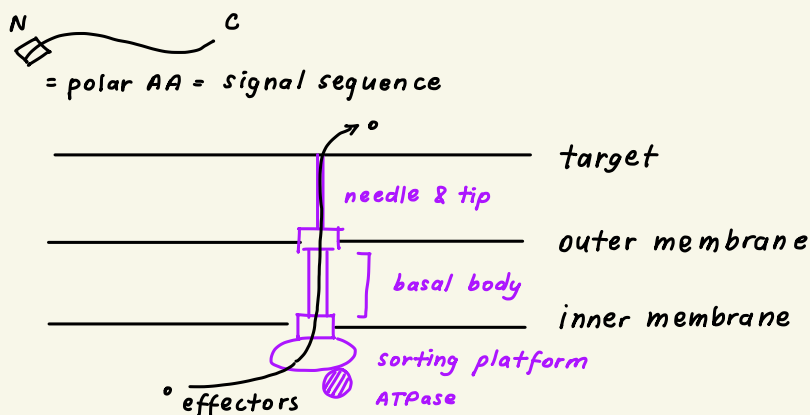
- humans are dead-end hosts

- blood meal enter fleas blood meals

- Yersinia travels to the gut upon consumption
- the yersinia moves to esophagus & blocks the throat
- repeatedly bite people to get a meal
- the blockage is suddenly released, causing the flea to vomit pestis into the bloodstream
- kills the flea

human bloodstream → infected lymph nodes → buboes = bubonic plague → actually a precursor

- lymph nodes burst open + stank
- miasma theory at the time dictate that this is caused by smell
- how do they infect cells & cause disease?
 - early intracellular lifecycle
 - late extracellular lifecycle
 - very good @ blocking phagocytosis
- macrophages act weirdly when infected
 - ↑ cell gets round, ↓ phagocytosis
 - 1990: express YopE protein
 - upon deletion, macrophages can then phagocytose yersinia
 - YopE is not sufficient on outside (must be injected cytosolically)
 - 1991: injecting recombinant, purified YopE results in rounding YopE
 - 1994-96:
 - contact b/w host & pathogen
 - YopE can be visualized getting delivered into the cell
- YopE is an effector protein
 - rewires some native host cell function
- T3SS first identified in Yersinia
 - encoded on a virulence plasmid (genetically identified)
 - 1998: first view in Salmonella
 - able to extract needle complexes
 - long extensions, syringe like shape
 - 25+ proteins that assemble & oligomerize
 - may also be embedded into chromosome
 - delivered as parts of pathogenicity island/alongside effectors
 - ALL are contact-delivered
 - yersinia has to sit on the surface
 - block signalling & polymerization to avoid getting eaten!



- only able to translocate unfolded proteins
- T4SS is extremely widespread
 - virulent & nonvirulent
 - proteins and/or DNA are substrates
 - signal sequence is not well-conserved
- 1990: T4SSa (Vir) discovered in agrobacterium tumefaciens
 - delivers oncogenic/proliferative DNA proteins to create tumors
 - energy, channel, pilus

- 1998: T4SSb (DOT/ICM)
 - discovered in *Legionella*
 - dot = defective for organelle trafficking
 - *legionella* is EXTREMELY broad
 - hang out in water system and many amoeba
 - secrete 300+ effectors
 - required to interact w/ so many amoeba
 - problematic to study b/c redundancy
- T6SS: found in pathogens & nonpathogens
 - toxic effectors that can potentially affect bacteria & eukaryotes
- first found in 2006: in 25% of gram $\ominus$ bacteria
- 13 different core subunits
 - very similar to a T4SS
 - need to deliver tail complex
 - look similar to phage proteins
 - acquired & flipped to enable deliver
- "spring-loaded poison dagger"
 - contractile outer sheath propels inner tube
- used for killing non-kin
- T7SS:
 - 2003 in *Mycobacterium* (ESX1-5)

- ① many proteins promote pathogen survival & disease
- ② elaborate on existing methods to enable delivery

7.26 Lec 11 - Cytosolic vs. Vacuolar Pathogens

- ① cytosolic vs. vacuolar lifestyle
- ② diseases and clinical outcomes
- ③ regulating pathogenicity

Listeria Monocytogenes:

- 6⊕, bacilli (rod shaped)
- first identified in rabbits, ID'd as human pathogen in 1920's
 - Soil/H₂O → Humans
 - food-borne
- 1982: 51 people
- hot dogs, ice cream, salad, soft cheese
- refrigerated, mass produced, UNCOOKED food
- can grow at 4° (in your fridge) - create a bolus
- would crop up in plants
- not pathogenic until it enters the body @ 37°C
- HIGHLY ADAPTABLE
- biggest outbreak: 2017/8 in South Africa
 - 200+ deaths
 - have to burn/bury this food
- globally = 23K/year, US = 1.6K/year
 - not very high
 - ~20% mortality rate

Listeriosis:

- if healthy, acute intestinal infection
 - dosage = 10⁹ listeria required
- immunocompromised/ pregnant
 - enter blood/lymph
 - go to spleen/liver
 - crossing blood brain barrier → meningitis
 - bacteremia/sepsis (accumulate in bloodstream)
 - cross placenta → fetal infection, miscarriages
 - only requires 10³-10⁴ bacteria
- we're not sure how its crossing these barriers

Listeria's Infectious Lifestyle:

- depicted in 1989 via EM
- listeria has virulence factors on its surface
- zipper invasion
 - hijack receptor-mediated endocytosis
 - something on pathogen recognize surface proteins
 - determines cellular tropism
 - Internalins: some polymerization
 - Internalin A: E-cadherin (epithelium)
 - Internalin B: cMet
 - Internalin F: Vimentin (brain barrier)
 - bad @ causing infection in mice b/c point mutation in E-cadherin
 - activation of actin polymerization to encourage endocytosis
 - pull a membrane over themselves
- vacuole escape
 - phospholipases - PlcA, PlcB
 - LLO (hemolysine, β-pore forming toxin)

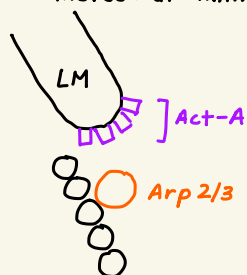
- put a bunch of β -PFT (inactive) on vacuole surface
 - docks on cholesterol
 - @ pH 5.6, LLO pore insertion is activated
- observe "waves of infection"
 - 1986: interfering with cytoskeleton
 - nocodazole = microtubules, cytochalasin D = actin

- actin tails

- filaments forming behind the listeria bacteria
- pathogen seems to be inducing this (start @ the poles)
- transposon screen to find actA
 - do a plaque screen
 - look for anything that prevents infection (actA)
 - reintroducing actA restores infection
- allows bugs to rocket around in cells

- actin-based motility

- molecular mimicry



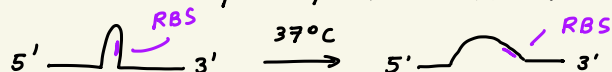
- activate polymerization w/ Act-A
- Arp 2/3 complex is the catalytic engine
- N-WASP is the controller in host cells
- Act-A is very similar to N-WASP (mimicry)

- stage-specific regulation:

- PrfA is the masterswitch
 - transcription regulator
 - target a variety of virulence factors

① Thermoswitch in mRNA of PrfA

- low temp. keeps RBS hidden (cant be accessed by



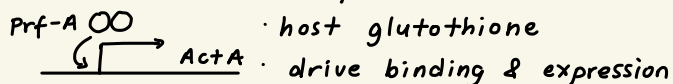
- opens up @ 37°C

- PrfA can activate internalins, Act-A

② Promoter Affinity

- high affinity activate Plc, LLO
- low affinity ActA

③ sense the host cytosol



Salmonella (Enterica)

- lives inside the vacuole
- different subgroups: divided into serovars
 - antigenic differences that change our ability to interact w/ it
 - capsule, flagella, outer PL layer of Gram $\ominus$ Bacteria

① Nontyphoidal (Salmonella)

- S Typhimurium
- gastroenteritis
- 1.3 billion cases/year
- animal reservoirs: poultry, reptiles, hedgehogs, pets
- small intestinal cells

- generalist pathogen
- once inside cells, trigger innate immune system
 - cytokines → inflammation
 - unpleasant but effectively clears the bacteria

① Invasive NTS

- systemic infection w/ co-infection, malnutrition, immunocompromise

② Typhoidal (S Typhi/Paratyphi)

- enteric, step-ladder fever (typhoid/paratyphoid), anorexia
- 10-20 million cases per year
- 10-20 million cases, 1% mortality
- specialists spread human to human
- antibiotics aren't very effective
- chronic carriers (0.1-3%)
 - asymptomatic
 - gastroenteric spread

ex. typhoid Mary

- salmonella virulence

- pathogenicity islands
- SPI1 = T3SS-1
 - support invasion
- SPI2 = T3SS-2
 - intracellular growth & systemic infection

- trigger invasion

- inducing phagocytosis
- totally mediated by T3SS
- create membrane ruffles
- mimic host signalling proteins
- Rho-GDP $\xrightarrow{\text{GEF mimics}}$ Rho GTP ON
- GAP mimics

↓
activation of Arc2/3
/actin formation

- sensing vacuole

$Mg^{2+} \uparrow$ $pH \uparrow$

① environmental cues

② 2 step signalling

· swap b/w SPI-1 + SPI-2

③ SCV moves to microtubule organizing center

④ insert into golgi + dock

⑤ force all vacuoles to provide membrane & nutrients to bacteria

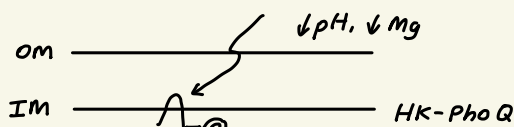
(can grow the vacuole + fill w/ listeria)

SIFs

hijack kinesin

SCV

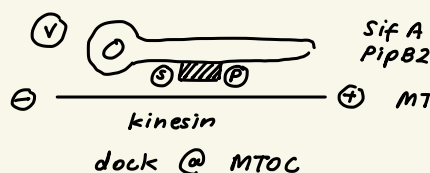
②



PhoP $\xrightarrow{\text{phosphorylation}}$ PhoP~P (Response Regulator)
↓
SPI-1 SPI-2

④ SCV SseF, SseG

golgi



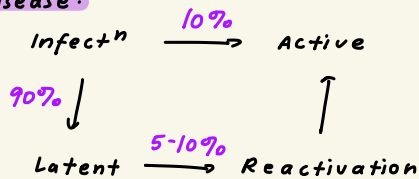
"stretching & stealing"

7.26 Lec 14 - TB Guest Lecture

- Bryan Bryson

- Initially, TB was thought to be inherited
 - multigenerational households prompt this hypothesis of etiology
- Koch's Postulates
 - ① organism always present
 - ② pure culture
 - ③ cause disease when inoculated
 - ④ re-isolate (& match organism w/ original)
- TB has a super-low mutation rate
 - DNA polymerase proof-reading mechanism

TB Disease:



- If this is true, might want to learn the difference b/w latent & active disease
 - "reactivation" of disease NOT the bacteria
 - immune system is just keeping the infection @ bay more effectively
 - latency is not so permanent
 - HIV, TNFs-blockers
 - metabolic health (TB as an immunometabolic disease)
 - #2 comorbidity is diabetes
- crude molecular differentiation b/w latent & active

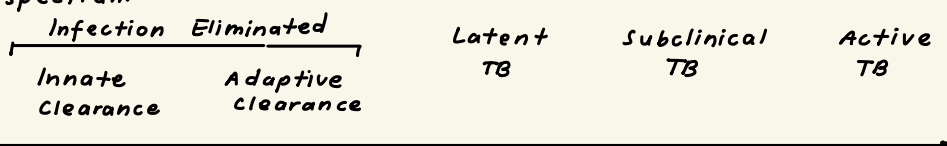
Life cycle:

- aerosolize bacteria & expel them
- only ~35 bacteria expelled per hour
 - all doors & windows are open
 - in TB endemic areas, many homes don't have windows
- enter alveolar space
 - resident immune cells patrol
 - 1 macrophage / 3-5 alveoli
 - time before innate immune cell recognizes is large
- granuloma
 - of innate immune cells
 - fibrose or necrose
 - necrosis break outer barrier → allow bacteria to exit

TB Disease (updated model):

- lung TB → extrapulmonary sites
 - granulomas rupture, bacteria enter the bloodstream, exit
 - liver, meningitis

- Spectrum:



- everything is much slower b/c no infective burden

- adjuvants push the immune response
 - extra stimulation

TB Diagnosis:

① skin test

- inject purified protein under skin
- check for immune response
- induration → need further analysis
 - some nonspecific mycobacteria proteins
 - BCG vaccine = guarantee rxn

② IGRA: Interfering Gamma Release Assay

- check T cell response to proteins from RD1
- need ELISPOT reader

③ Ab-based diagnostics

TB Treatment

- prior to antibiotics
 - sanatorium
 - industrialization means extra packed working conditions
 - collapse therapy
 - try to collapse the lung??
 - fresh air & sunshine
- antibiotics: initially given as monotherapies
 - streptomycin
 - isoniazid → mycolic acid
 - rifampin → polymerase
 - nowadays + 2 more drugs
 - ethambutol
 - pyrazinamide → mech. unclear
 - nowadays, various drug resistant TBs exist
 - 2019 discovery of very metabolically toxic but effective antibiotic

Mycobacterium Tuberculosis (Mtb)

- hypothesis: all TB originated w/ soil bacteria
- Mycobacterium mycroti
 - rodent adapted
- not globally monolithic
- M. bovis
 - found in unpasteurized milk
- aerobic, non-motile, non-spore forming bacillus
- lots of mycolic acid
 - high molecular weight lipids
 - 250 genes for lipid biosynthesis
- cigar band experiment
 - coat bacteria w/ amine reactive dyes
 - TB grows very polarly (from the poles)
 - poles don't grow @ same rate (asymmetric)
 - don't divide down the middle
 - progeny have different antibiotic resistances

- form of natural heterogeneity

- 800 long mycolipids

- a lot of hydrophobicity in the cell wall

- the whole colony must be acquired

- TB is so waxy it can slide

- pull agar along w/ colony

- extremely hydrophobic cell wall makes drug discovery difficult

- intracellular lipid occlusion

7.26 Lec 16- Immune Systems: Bacteria

- 1) phases of host defense
- 2) sensing mechanism
- 3) evasion mechanism
- 4) Contribution to disease

Host Defenses

① Barriers

- skin, pH in stomach, mucus, tears, saliva
- physical/enzymatic (lysozyme for bacteria)
- microbial barriers (commensals)
- nutritional immunity
 - limiting iron access
- many points of attack to deal with many evasion techniques

② Innate

- phagocytosis
- inflammation
- antimicrobial peptides
 - most ancient immune mechanism
 - first line defense
 - secreted by host cells
 - defensins:
 - cationic to interact with highly anionic bacteria
 - oligomerize to form pores
 - bacteria do electrostatic repulsion by making their proteins more positive (staph aureus)
 - anthrax releases proteases to destroy peptides

③ Adaptive

- antibodies
- T cells

Complement System:

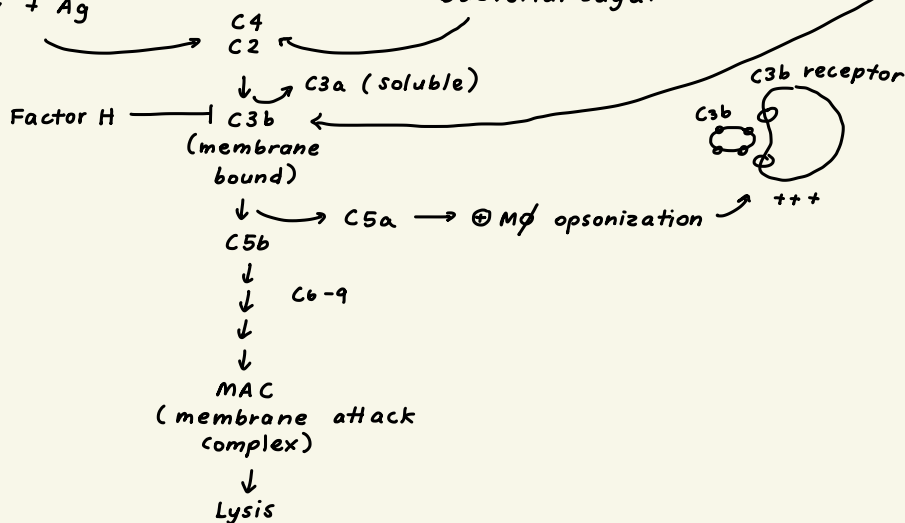
- soluble protein cascade
- blood proteins

classical

antibody + Ag
C1

lectin
bacterial sugar

Alternative



① Lysis via MAC

② Opsonization

- C3b
- assist in recognition by macrophage

③ C3a → Inflammation

- activate + recruit sites
- factor H decoration on host cell to prevent unwanted lysis

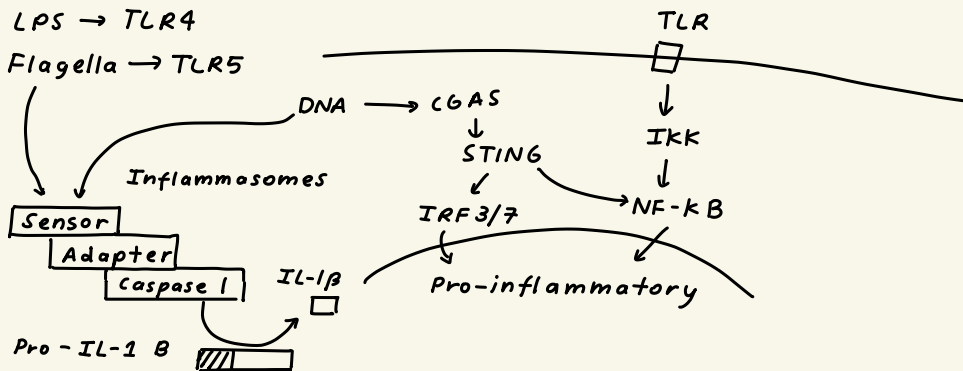
Complement Evasion

- change capsule
 - capsule = extended polysaccharide coating
 - recruit factor H
 - hide C3b from immune cells
- Gram $\oplus$ $\rightarrow$ MAC b/c peptidoglycan
- cleave complement
 - EHEC $\rightarrow$ Serine protease P
 - EHEC $\rightarrow$ T5SS Pic
- becoming intracellular means most populatⁿ won't see complement

PAMPs / DAMPs $\rightarrow$ PRRs

- LPS $\rightarrow$ TLR4

- Flagella $\rightarrow$ TLR5



Evasion Tactics:

① Disguise

- *S. typhi*
 - thick cover (capsule) that covers LPS
- glycosylate flagella proteins to stop TLR4
- yersinia uses effectors to block receptor

② Remove Stimulus

- particularly in the host
- *Listeria* @ 37 $^{\circ}$ $\rightarrow$ flagella
 - b/c they can just hijack actin when they're not @ 4 $^{\circ}$ (in environment)

Phagocytosis

Complement



- needs to eventually undergo lysosome fusion
 - $\downarrow$ pH, enzymes
 - phagolysosome
 - ROS (reactive oxygen species), NO
 - Siderophores
- secrete cytokines + inflammation
- present pMHC II $\rightarrow$ CD4

Evasion Tactics:

① Block uptake

- capsule can be a great way to block complement receptor on phagocyte
- pore forming toxins specific to immune cells

② Block phagolysosome function

- Virulence Factor $\rightarrow$ V-ATPase ($\downarrow$ pH)
- siderophores to outcompete iron
- ROS production

③ Escape to the cytosol

- listeria escape to cytosol w/ LLO

Adaptive Immune System

① Antibody / B cells

- neutralization (prevent normal targeting)
- opsonization (covers it in Ab)

② CD8 T cells

- pMHC I $\uparrow$

③ CD4 cells

- helper T cells
 - release cytokines that act on a variety of cells

$(T_H1) \rightarrow IFN\gamma \rightarrow \oplus M\phi/DC$

$(T_H2) \rightarrow IL-4 \rightarrow \oplus B \text{ cells}$

Adaptive Immunity Evasion:

① Secrete proteases to degrade Abs

- IgA protease

② $\downarrow$ MHC (MHC I)

- helps hide

- these cells are all migratory

③ Block T cell $\oplus$ / Proliferation $\rightarrow$ Salmonella $\rightarrow$ SseI $\rightarrow$ DC migration

- hyperactivation of immune cells

- granulomas
- clots

- sepsis

- systemic inflammation
- entire body is overreacting

① sepsis

- fever, high $\heartsuit$ rate, elevated breathing

② severe sepsis

- organ failure

③ septic shock

- inflammation destroys vasculature $\rightarrow$ low BP $\rightarrow$ 30-50% fatality
 - normally, cytokines locally modulate vasculature to let some immune cells into tissues
- septic patients have elevated c5a levels
 - make immunotherapies (make c5a Ab's)

19 million
cases/
year

7.26 Lec 17 - Microbiomes + Host Health

A. What do commensals do? Do they matter?

- 20K genes in a person
 - each person has a microbial metagenome that is 100-1000x larger than human genomes
 - b/w 2 unrelated info, there is >90% uniqueness in metagenome
 - may explain differences between characteristics of individuals
- 2015 paper: wild microbiome protects against lethal influenza infection in mice
 - first paper to indicate the importance of the microbiome
 - effect size is very large in mice!
- ... protects against cholitis induced colon cancer
 - all mice got cholitis but microbiome immune modulation prevents the development of cancer
- 2018, science: humans got cancer immunotherapy
 - most people generally don't respond
 - patients who did respond had distinct gut microbiomes
 - transplanted microbiome into gut → sufficient to protect mice
- paternal microbiome perturbation impacts offspring fitness
 - either deplete microbiome or not
 - allow some to recover, some to not
 - mate w/ naive female mice
 - compare fitness of F1 progeny : pups are small, improperly developed, fail to thrive
 - clearly the microbiome is really important in all phases of development
 - epigenetics?

B. Host Landscape + Microbiome Composition

· Host Features

① Diet

- bacteria are totally dependent on you to feed them
- made modifier

② Moisture

- profound effect
- gut is very moist, skin is very dry (by comparison)
 - skin only has only gram positive bacteria
 - b/c of the thicker peptidoglycan layer

③ Salt content

- most mucosal surfaces are low salt
- skin is salty - peptidoglycan help deal w/ osmotic stress

④ pH

- low in stomach + skin → inhibit pathogen colonization
- neutral elsewhere
- restricts what is able to survive

⑤ Movement

- gut peristalsis, skin is stationary
- bacteria in gut have flagellar
- all skin microbiota are nonmobile (don't illicit TLR response)

⑥ Oxygen

- skin is aerobic except deep in hair follicles
- colon is mostly anaerobic
 - but all famous gut pathogens are aerobic?
 - infection requires some gut disruption

⑦ Host Immune Response

- don't completely understand relationship b/w this and immune system
- in the gut, bacteroides causes a gut IgA response
 - helps increase colonization
 - w/o IgA coating its harder for them to stick to the gut
 - coating doesn't disrupt the bacteroides life cycle!

· microbe-microbe interaction

① Competition for nutrients (metabolic niche)

- result in "priority effect"

② crossfeeding

- Bifido Bacterium secrete sialidases which produce sialic acid and oligosaccharides prompting growth of bacteria that can utilize those nutrients

③ physical competition

"colonization resistance" prevent pathogen invasion

④ direct inhibition of other microbes

- produce small molecules to kill competing bacteria
- cutibacterium acnes (skin commensal) produces cutimycin that inhibits staff aureus

⑤ cross-species quorum sensing

- → autoinducing peptide quorum sensing peptide small amount!

- • • → ↑↑↑↑ sensed by sensor (protein receptor) kinase → TF → genes upregulated

- sense the number of "siblings" that are nearby
 - help bacteria determine how to modulate their behaviors

⑥ phage predation

- viruses that infect bacteria
- trying to use phages to modulate microbiome composition

C. How do we know the members of the microbiome?

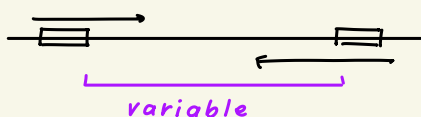
- Methods to identify members:

① culture

- put samples on bacterial membrane
- poor technique b/c media is very different from gut bacteria
- now, we also use horse blood for "mammalian nutrients"
- low throughput and biased
- major advantage: can study the microbe
 - we're great @ sequencing but how do you keep investigating?

② 16S sequencing

- universal primers bind conserved sequences



- extremely low biomass and use these primers to PCR amplify and then sequence
- compare to reference database
- caveat: totally dependent on reference database
 - no information outside of PCR sequence
 - level of genus but species/strain is harder

③ Metagenomic Sequencing

- complete view of genes in a particular microbiome sample
- provides a very fragmented view
- still depends on a reference database
- computationally tile genome fragments to try to match an existing bacterial genome
- much more info about functional genomics
- really hard to do causation experiments in this context

④ metatranscriptomic sequencing

- sequence cDNA
- actually sequencing what gets expressed
- even w/ identical metagenomic sequence, can have very different metatranscriptome
- hard b/c bacteria are so much smaller than human cells
- RNA is unstable relative to DNA
 - likely cDNA = bacteria was alive @ time of sequencing
 - for metagenomics, many sequenced DNA samples are dead

D. what defines a commensal vs. a pathogen?

- contextual pathogenicity (exist on a spectrum)

ex. *staphylococcus epidermis*

- generally a commensal
 - colonization resistance
 - promotes skin Treg in early life
 - one of the first commensals in babies
 - immunosuppressive
- can ALSO cause disease
 - neonatal sepsis
 - determined by host factors
 - premature babies, entering the bloodstream
 - device infection
 - atopic dermatitis

ex. *staphylococcus aureus*

- generally a pathogen
 - infection/eczema
- BUT common in nares
 - provide a protective antibody response against infection
 - interxn w/ *corynebacteria*
 - suppresses virulence genes

- comparing features

· commensal features

- ① exist within a community
- ② long term colonization
- ③ intact barrier → surfaces good @ microbe interxn
- ④ no inflammation
- ⑤ some tolerance (good!) ← similar →

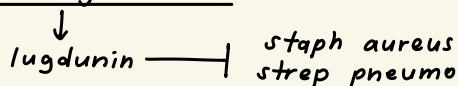
· pathogen features

- ① 1 bug dominates
 - ② most infections are acute
 - ③ infection in normally sterile sites → blood-stream
 - ④ cell injury + inflammation
 - ⑤ immune evasion
- but context is important!

D. How do microbiota affect infectious disease?

- microbe-microbe interxn

· *staph lugdenensis* (commensal)



- 6-fold lower rate of co-infection w/ these bugs

- why isn't everyone colonized?
- require siderophores but not encoded by its own genome
 - has to be created + shared by other commensals
 - must create a synthetic microbiome
- host Interxn
 - modulation of host immunity
 - ① non-antigen specific effects
 - microbiome "educates" host immune cells
 - gut microbiome → cancer immunotherapy
 - ↘ vaccine efficacy
 - antibiotics → worse outcomes
 - no shared antigens b/w microbiome + vaccine or microbiome + tumor
- ② antigen-specific effects
 - some strains of *S. epi* are good @ inducing CD8⁺ T cell response
 - create strains that generate tumor-specific T cells to treat cancer!

7.26 Lec 18 - Opportunistic Fungal Pathogens

Opportunistic Infections:

- many organisms can exist as a commensal
commensal $\xrightarrow[\text{Immune System}]{\text{T}}$ pathogen
- mostly caused by immune breakdown
- immunocompromised
 - age, immunosuppression, chemotherapy
 - antibiotics reduce the microbiome
 - more conducive to fungal pathogen
 - coinfections
 - HIV, malaria

Candida Albicans:

- 30-70% of healthy people
- leading cause of bloodstream infections
 - immune suppression
 - medical devices: can exist on plastic surfaces
- causes candidiasis
 - ① mucosal = mouth = thrush
 - white patches of yeast in the mouth
 - particularly common in children
 - = vagina = yeast infections
 - frequently caused by hormone changes
 - menopause, diabetes
 - ② invasive/systemic
 - brain, ♡, bones
 - 35-60% mortality
 - enabled by how malleable this organism is

Morphology Changes:

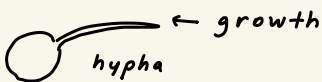
- pleomorphic fungi

① budding yeast



② pseudohyphae: clumps of yeast cells

③ filamentous



- both budding + filamentous are pathogenic!
- filamentous is particularly good @ tissue invasion
- budding help w/ initial adhesion + dispersal
- cycle b/w these two states

- healthy:

- really low fungal load
- mostly in the yeast form
- primarily exist in mucosa

- immunocompromised

- high fungal load, ↑ filamentous
- invade bloodstream + tissues

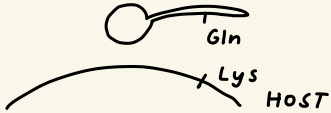
Cell walls Glycoproteins:

- thick cell wall
 - polysaccharide chitin } **PAMPS**
 - mannins } → complement activation
 - exploited as immune targets
- support adhesion
 - extracellular pathogen
- ① Agglutinin-like Sequences



- bind ECMs, cadherin
- promiscuous protein binding target
 - c terminus
- biotic + abiotic targets
 - enables medical device infection
- help them bind to each other

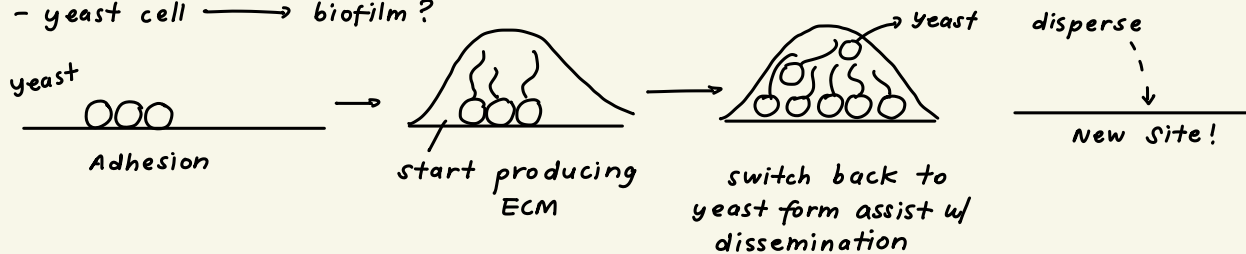
② hyphal wall proteins



- enzyme in body
 - forms covalent bonds to ECM

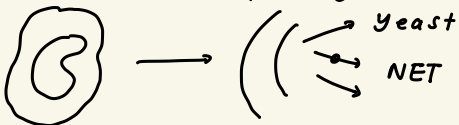
Biofilm Formation:

- fibrous coating
- yeast cell → biofilm?



Scaffold for Growth:

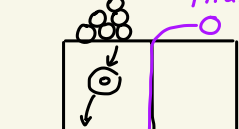
- resist antimicrobials
 - ? ↓ diffusion
 - ? bind drugs
- resist immune system
 - can also kill pathogen w/ neutocis



- neutrophils literally explode

Colonization → Invasive Disease:

yeast filamentous ① Early: induced endocytosis



ALS3 → cadherin → clathrin mediated endocytosis

- transphagocytosis?
 - transfer from apical → basal membrane

② Late: Hyphae Penetration

- between + through cells
- barrier disruption
 - cleave tight junction proteins

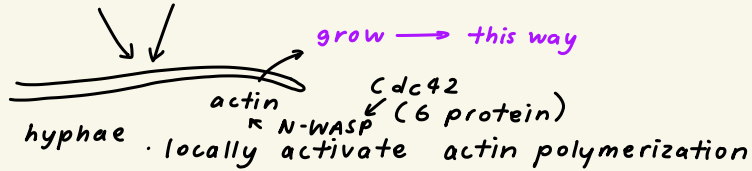
bloodstream

Secreted Aspartic Proteases: fungal secretion

- ① Degrade cell-cell junction components
 - connective tissue
- ② Penetrate mucus lining mucosal surfaces
 - help increase chances of adhesions
- ③ cleave complement receptors
 - limit detection & clearance
- ④ nutrient acquisition
 - enable nitrogen acquisition

Regulate Morphology:

- 37°C, Blood (low glucose)



- Farnesol
 - quorum sensing molecule
 - inhibits hyphae formation
 - increase yeast form to increase dispersal

cell wall composition:

- Mannans → TLR4
- highly variable
 - niche specific adaptations
 - hypha → ↑ chitin → Mφ recognition

Resisting Phagocytes:

- ① SAP → Phagocytosis
- ② Block ↓ pH
- ③ size
 - size extremes can limit phagocytosis
 - > 20 μm → Phagocytosis
 - some yeast cells are huge
 - titan cells
 - spherules
 - hyphal form of candida can be problem
 - phagocytosis takes ~2 hrs
 - hyphae can grow 18 μm/hr
 - simply outpaces the macrophage
 - "frustrated phagocytosis"
 - some immune cells fold the fungal hyphae
- ④ pyroptosis
 - not special to bacterial cells
 - ⊕ Inflammasome
 - ↳ lytic cell death
- ⑤ Lytic Escape
 - eventually
 - the hyphae stretch the
 - membrane so much it breaks



⑥ vomocytosis

- nonlytic exocytosis
- not clear how this is happening?

Virulence:

- Δ in morphology
- SAPs
- variable lifestyle in different niches

ENVIRONMENTAL FUNGAL PATHOGENS

Topics:

- 1) *Histoplasma capsulatum*
- 2) Coccidioidomycosis
- 3) Zombie-making fungal pathogens

I. *Histoplasma capsulatum*

- The most common cause of fungal respiratory infections in immunocompetent hosts
- Why is it called 'Cave disease'?

spores are commonly found in the bird & bat droppings found in caves

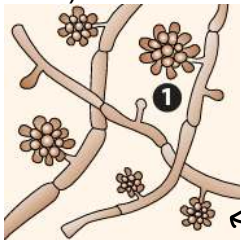
- Histoplasmosis:
 - Pulmonary
acute respiratory disease similar to pneumonia!

- Disseminated
"metastasis" of fungus; spreads from lungs to other organs
more common in immunocompromised individuals!

Thermally dimorphic fungi

- hyphae produce asexual spores named microconidia or macroconidia based on size

1) Environmental form



grow as hyphae
@ 25°C

vs

2) Pathogenic form



yeast form in host
tissues (@ 37°C)

spores called
conidia

- microconidia (small, infectious)
- macroconidia (large, with distinctive projections)

Regulation of morphology

- temperature dependent gene regulation
- RYP1-3 specifically involved in yeast-phase induction

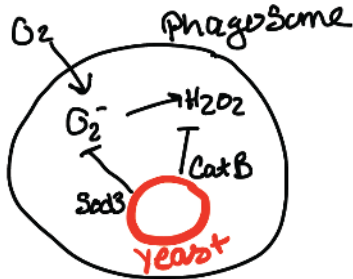
Intracellular lifestyle – what strategies enable this lifecycle?

1) Shielding: (common defense strategy for many yeast)

polysaccharide capsule-like coating w/ altered surface antigens to enable immune evasion

2) Countering

α glucans to shield β glucans (which are recognized by Dectin 1 receptors)



produce enzymes (catalase, superoxide dismutase) to neutralize reactive oxygen species within the phagosome + evade degradation

3) Coping

(acid tolerance response)

survive in acidic phagosomes

ATR

& nutrient acquisition mechanisms

(siderophores)

4) Dissemination via host apoptosis

can induce apoptosis in host macrophages aiding in release + spread to other tissues



5) Granulomas

formation of granulomas helps the host contain the infection but also provides a niche for latent fungal survival

II. Coccidioidomycosis (Valley Fever) *↪ actually transmitted via dust storms!*

- Coccidioides spp. are endemic in the southwest region of the United States (U.S.) and in other semiarid areas in Mexico and Latin America
- Caused by two species - *C. posadasii* and *C. immitis*
- Acute pneumonia
1-3 weeks after initial inhalation
flu-like symptoms, fatigue, chest pain, cough
rash (erythema nodosum)
- <5% transition to chronic disease
nodules & cavities
(akin to a TB infection)
- <1% develop Disseminated Coccidioidomycosis
fungus spreads to skin, bones, CNS (meningitis)
particularly common in pregnant women, the immunocompromised,
and people of African/Filipino descent

Dimorphic fungus:

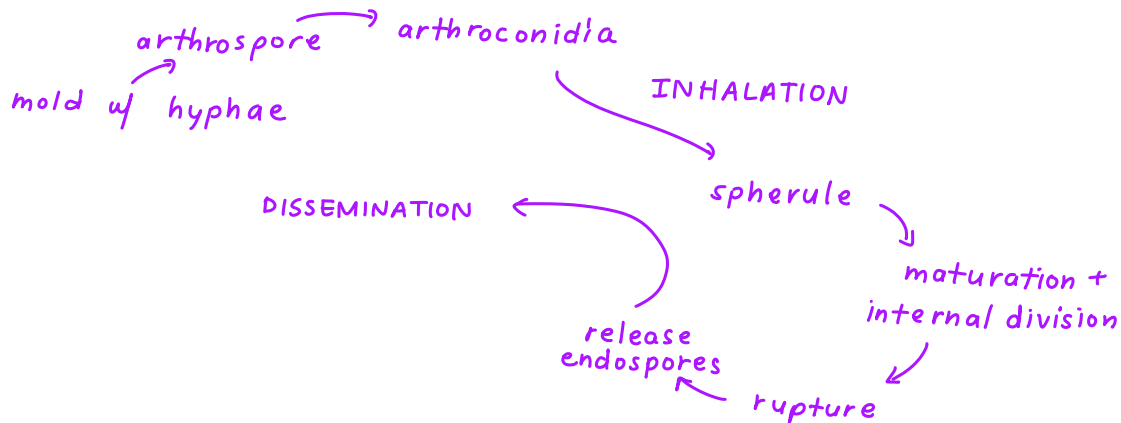
Mold/mycelium form → release spores known as arthroconidia (*barrel-shaped + infectious*)

Inhaled spores grow into spherules filled with 100's of endospores (*highly infectious*)

Endospores can revert to mold form in environment or make more spherules in host

large, thick-walled structures

coccidioides is also thermally dimorphic!



Mixed lifestyle inside host

- Large Spherule with Spherule outer wall glycoprotein (SOWgp) adheres to endothelium
help w/ endothelium adherence + immune evasion
- Released endospores (small) *and spread locally / to other organs*
- Later infection: As *Coccidioides* sporulates, it secretes MEP1 and they increase in size
MEP1 is a metalloproteinase that degrades host tissues facilitates endospore release + enlargement

III. Zombie-making fungal pathogens

- pathogen-adaptive behavioral alterations – examples and what's the ultimate goal?
manipulates host behavior to maximize its own reproductive success & dispersal

Ophiocordyceps unilateralis

- “Zombie ants” and the death grip
 - infects ants (specifically carpenter ants)
 - cause ants to climb up vegetation + “death grip” (bite down very hard)
 - fungus kills the ant & sporulates from corpse
 - increase dispersal via elevation

Entomophthora muscae

- Infection of house flies
- Summit disease
 - similarly, infected flies climb to high elevation & extend their wings
 - die in an optimal position for spore dispersal
 - eject to nearby hosts

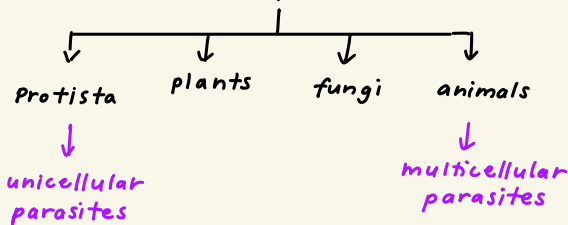
Lec 20 - Parasites

Parasites:

- eukaryotic
- organisms that require host for nutrition/ replication
- detrimental to host
- 3 problems:
 - ① Establish niche in host
 - ② manipulate host factors
 - ③ decisions that help w/ replication vs. transmission

Classifications of Parasites:

Eukaryotes



- parasitism has evolved hundreds of times (not a clade)
- heteroxenous vs. homoxenous
 - heteroxenous = multiple hosts to complete the life cycle
 - ex. malaria = mosquito + humans
 - homoxenous = complete life cycle in 1 host
 - tropism = specificity of the host cell or organism
 - these concepts are related but not synonymous
- definitive vs. intermediate host
 - only relevant for heteroxenous parasites
 - definitive = required for sexual reproduction
 - intermediate = used for asexual reproduction
 - ex. malaria - human (intermediate) + mosquitos (definitive)
- short term cyclical phenotypes vs. long term disease phenotypes
 - short term = cycles of parasite development + replication
 - periodic fevers (as seen in malaria)
 - long term = due to chronic immune activation/infection
 - immune systems is much worse @ clearing parasites
- major classifications:

protozoans (1 cell)

metazoans (multicellular)

① insect vectors (heteroxenous)

- malaria - adapt to insect + human niche?

- infects ~20% people in Africa
- 2012: 627k deaths worldwide
 - 2021: COVID kills 400k people
- drug resistance is a big problem

- trypanosomes:

- African vs. American types
(tsetse fly) (chagas bugs)
 - 10k deaths annually from chronic ♥ disease
 - after decades of infection
 - the ♥ muscle becomes necrotic
 - no medications
- also affect other livestock

① insects

- fleas
- lice
- ticks
- mites (demodex)
 - cause problems w/ overgrowth

② helminthes (worms)

- most common parasite in developing countries
- direct infection
 - metabolic dysfunction
 - developmental delay
- decrease vaccine efficiency
 - really problematic for children

② contaminated food + water

- toxoplasma

- main vector is cat feces
- 30% seroprevalence (Ab)
 - not cleared by people
- no drug for chronic infection (cyst form)
- reactivates upon immunosuppression

- cryptosporidium

- cause diarrheal disease
- >30% in under 5 y/o children in Africa/Asia
- 1 approved drug in the US
 - only 75% effective

- giardia + entamoeba histolytica

- affect 20-40% population in developing countries

- commensal protozoa

- present in ~30% of healthy people
- purpose is unclear
- do they provide any extra immunity?

- flatworms (platyhelminthes)

· tapeworms (cestodes)

- undercooked meats = vector
- more common post-agriculture b/c we started eating livestock

- roundworms (nematodes)

· strongyloides

- soil contact w/ skin
- 800+ million global prevalence
- chronic hives + allergic skin/organ manifests

- most people are infected w/ multiple parasites @ once

- major developmental delays

- protozoans are single cell but often relatively large

- some worms cause lymphatic filariasis

- mites may cause rosacea

why do we study parasites? what makes them interesting?

- have shaped humans:

· physiology:

- malaria in Africa has positively selected for sickle cell trait
- sickle cell anemia - can't carry O_2 normally (pain + infection)
- allele is prevalent b/c it ↓ likelihood of malaria infection

· immunity: Type I (antiviral), Type III (extracellular, fungi)

- discovery of Type 2 Immunity

- allergies, asthma, eczema
- IL-4, IL-5, IL-13
- eosinophils, mast cells, hives
- positively selected for by historical parasitic rxns

· history:

- 1927 Nobel Prize: malaria therapy

- purposeful infection w/ malaria to create high fevers to clear syphilis
- syphilis is lethal but this has a 27% clearance rate
- treatment until penicillin in 1940's

- therapeutics are extremely scarce

· no vaccines until 2021 malaria vaccine

- only 60% efficacy

· helminths are extremely common

- only 6 approved drugs for all of them
- cancer has Jan 2023 - 2024 25 new drugs
- research efforts are lopsided

· chronic disease

- huge economic impact
- development delays

- parasites are eukaryotic

· H16H coding capacity

- malaria genome: 23 Mb

- whereas HIV is only 9 kb

- large diversity of lifestyle adaptations + methods to avoid immune system

· hard to target essential cellular processes

ex. in trypanosomes:

- thousands of coat proteins (VSG)

- needed for adhesion/physiology

- only expresses one @ a time

· similar to T & B cells

· enable coat protein switching to evade immune response

- have a similar method to VDJ recomb

· 70 bp conserved sequence upstream of VSG genes

· occurs in 1/10⁴ cells

· triggered by ds breaks in genome

· perfect cadence to continually change antigenic target

- how does a trypanosome sense its neighbors?

· independently evolve quorum sensing system

① proliferative form (slender)

- autoinducing C1 peptides

- sensed by GPR84

- trigger differentiation into "stumpy form"

② Stumpy form

- G1 arrest (for transmission)

- downregulates VSG

- ↓ VSG expression (can't Δ coat protein anymore)

- protects parasite from proteolysis in the fly gut

· switch required for transmission vs. replication

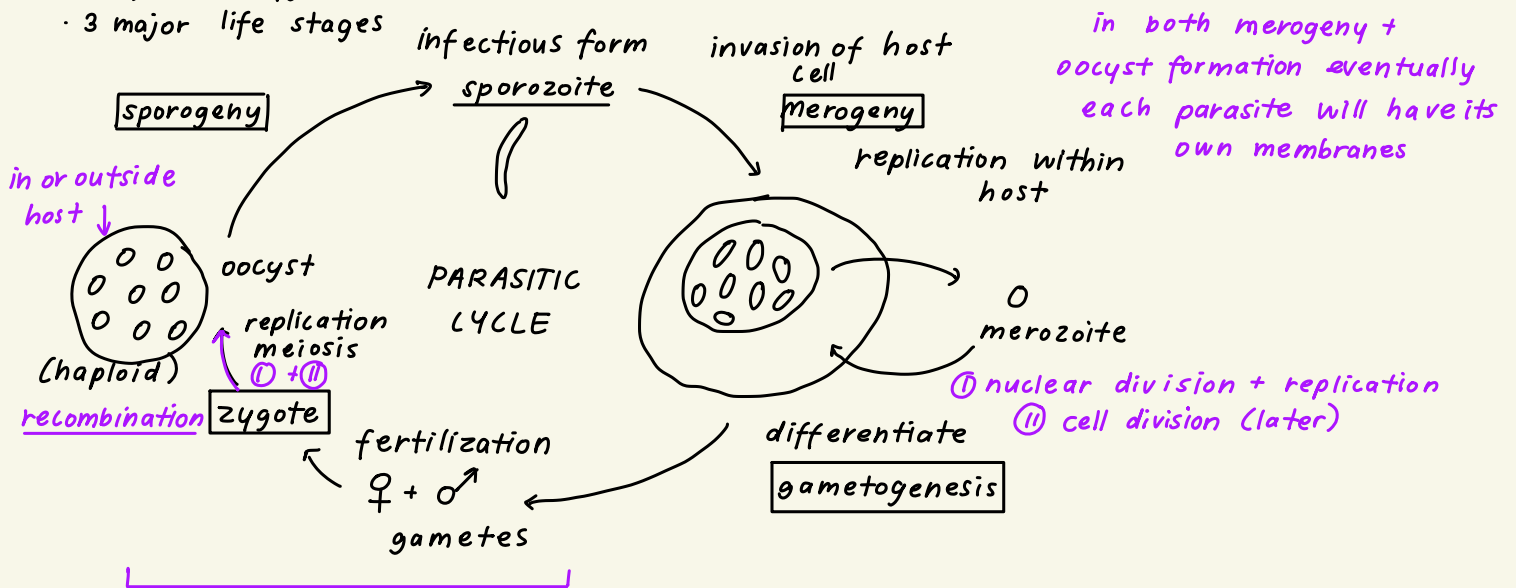
· what abt fly immune systems?

- they don't have adaptive immune systems!

7.26 Lec 21 - Malaria Pathogenesis

Apicomplexans:

- protozoa
- all are parasites of animals
- 3 major pathogens:
 - ① Plasmodium: Malaria
 - ② Toxoplasma (gondii): Toxoplasmosis
 - ③ cryptosporidium: diarrheal diseases
- have apicoplast
 - own 35 kb chromosome
 - nonphotosynthetic but looks like a chloroplast
- similar in size to NK cells
- extracellular + intracellular life stages
 - intracellular required for replication
 - travel extracellularly
 - enable different host cell invasion
 - has motile elements
- apicomplexans are haploid most of the time
 - only the zygote is diploid
 - 3 major life stages



- patterns of asexual reproduction
 - nuclear division/chromosome replication not linked to cell division
 - endodyogeny, schizogeny, endopolygeny
 - no real symmetry - how to sort nuclei (requires complex machinery)
- heteroxenous or monoxenous
- all human Plasmodium are transmitted by the anopheles mosquitos
 - importance of insect vector
- species of Plasmodium
 - in endemic areas, many people are coinfectd
 - confuse the immune system
 - enable genetic recomb for parasites
 - ① P. falciparum = 95% malarial deaths, dominates in Africa
 - ② P. vivax = most common outside Africa
 - ↳ develop @ lower temps
 - ↳ form dormant hypnozoites in the liver (non-rainy season) → can be reactivated!

- 2 ways that plasmodium affects humans

① selection for genetic traits @ evolutionary timescales

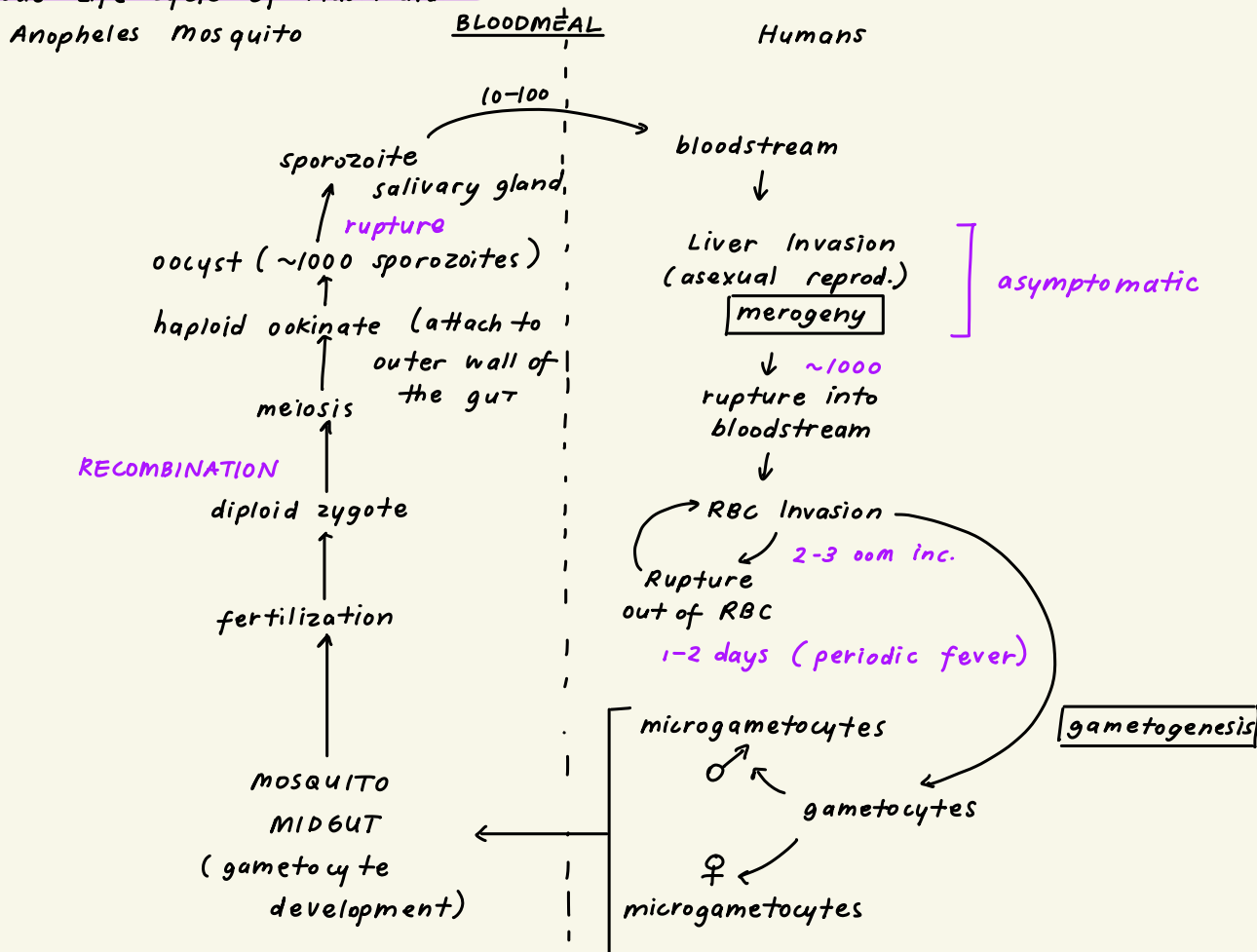
- sickle cell
 - 1 valine alters hydrophobicity
 - makes clumping easier
 - stalls plasmodium replication
 - sickling cause higher clearance in the spleen
 - increased clearance of plasmodium
 - homozygosity = sickle cell anemia

- plasmodium is particularly ancient

② Infection-induced disease

- mostly b/c RBC sequestration
- RBC clump → *b/c parasite division in the RBC* → cleared by spleen
 - ↳ occlude small vessels
 - ↳ strokes, kidney failure
 - anemia → respiratory distress
- more virulent strain simply b/c more RBC clumping

Infectious Life Cycle of Plasmodium:



sex is determined epigenetically

fate is flexible

(@ low density: more males than female)

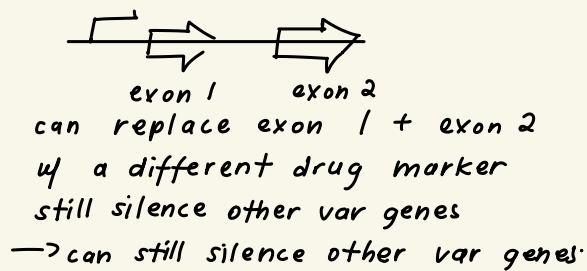
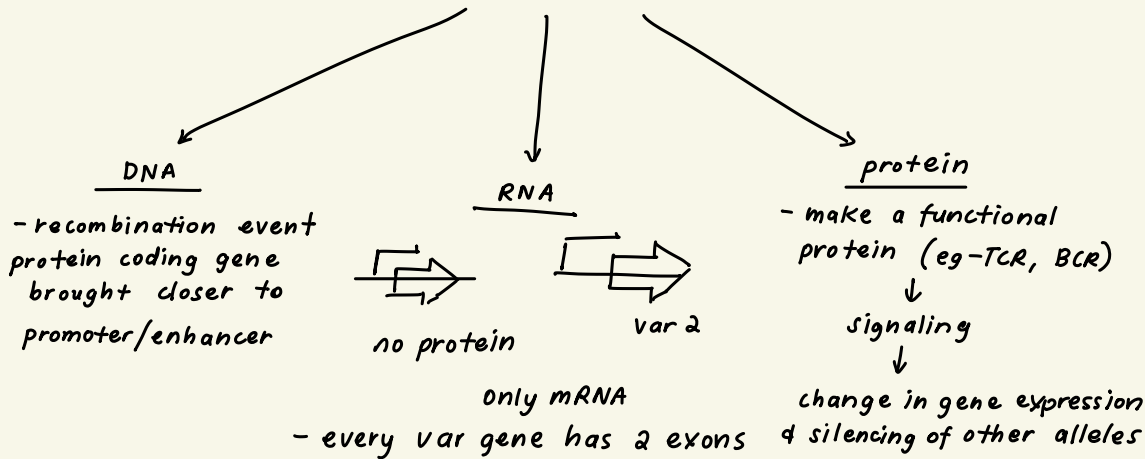
this doesn't seem to be reversible?

P. falciparum

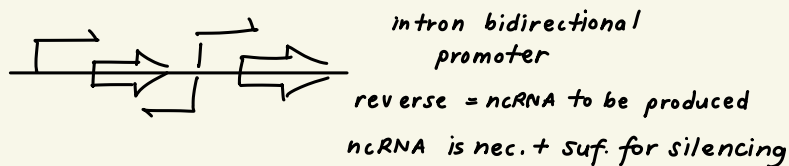
- lethality b/c RBC clumping
- express PfEMP1
 - unique to most virulent strain
 - gets exported to RBC surface
 - leave parasite cell membrane
 - leave parasitoid vacuole
 - reach ER
 - get secreted to RBC surface
- help RBC stick to endothelial cells
 - clumping inside blood cell
 - prevent spleen clearance of parasitized RBC
- very immunogenic
 - target of malaria vaccine
 - 65 PfEMP1 genes in each plasmodium
 - express only 1 var gene @ a time and switches b/w them
 - arsenal of many genes w/ slight variation

PK6 activation

↓ phosphatidylserine flipping in apoptotic host hepatocyte,
↓ phagocytic capture
(↓ immune response)



→ just take promoter + remove intron (no silencing) then relevant silencer is in the intron !!



- you can simply put an ncRNA on a plasmid + constitutive promoter to + silence all var proteins!

- seen in Barr Bodies

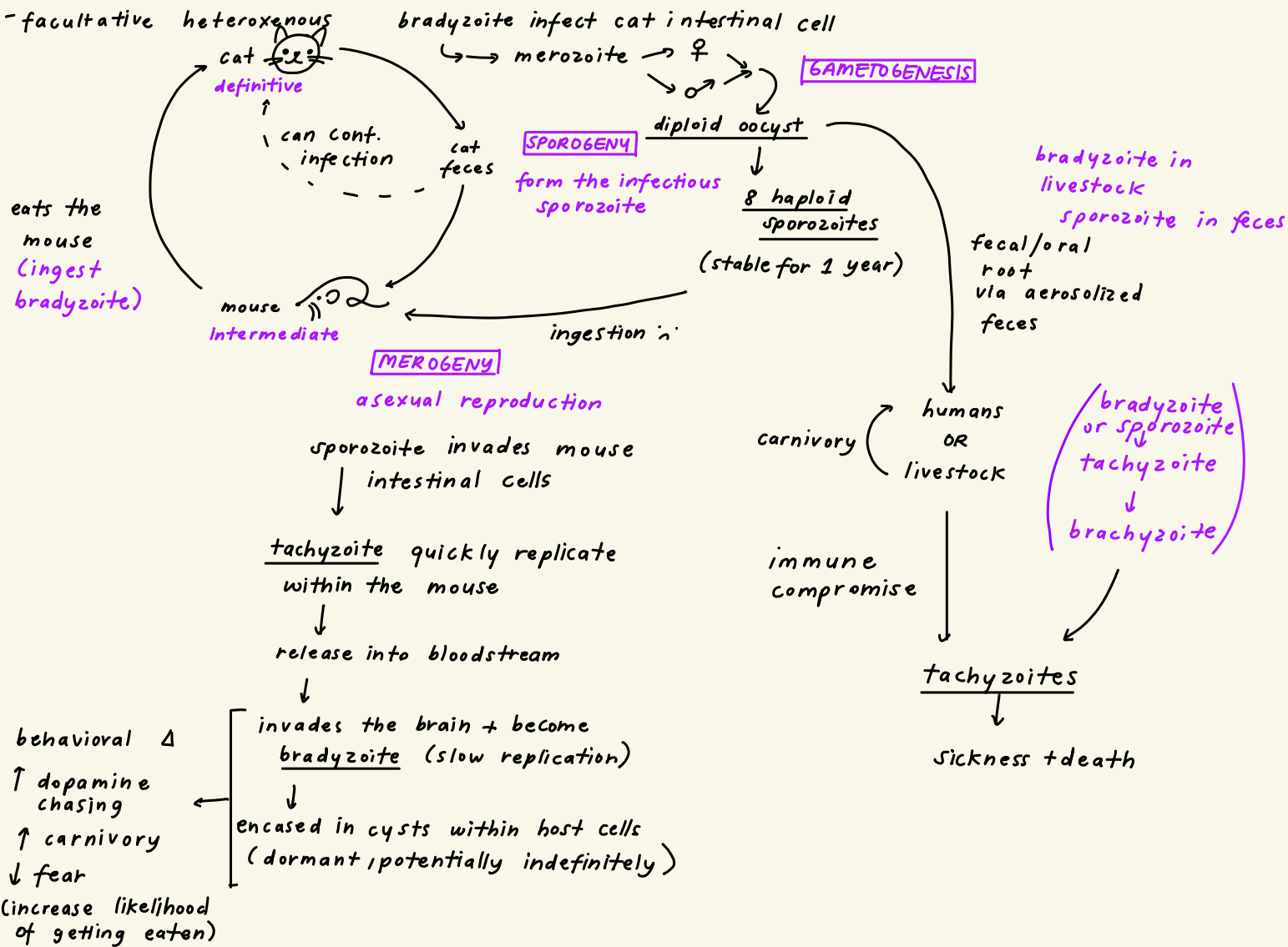
Xist ncRNA recruits epigenetic silencer

7.26 Lec 22 - Toxoplasma

A. Life cycle of toxoplasma

- affects behavior in mice
 - lose fear of bobcat urine
 - less fear, more carnivorous behavior, more risk-taking

- facultative heteroxenous



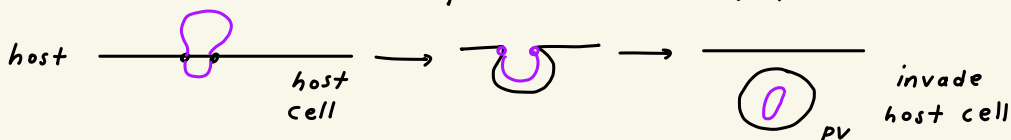
- antiparasitics work on sporozoite/tachyzoites

- nothing works on bradyzoite
- usually don't cause problems
- can be vertically transmitted to babies
 - cause pregnancy loss
- humans are an intermediate host
 - dead end intermediate
 - no evidence of humans being able to transmit the disease
 - does not drive toxoplasma evolution
- fecal-oral is the main means of intermediate host infection
 - 1 bradyzoite can infect a cat, 1k required for another mouse

B. Host Response to Toxo

- parasite entry

- parasite create a protein complex called a moving junction
- exclude host membrane from membrane of parasitoid vacuole



- exclude PRRs, components of immune system → parasite replicate in peace

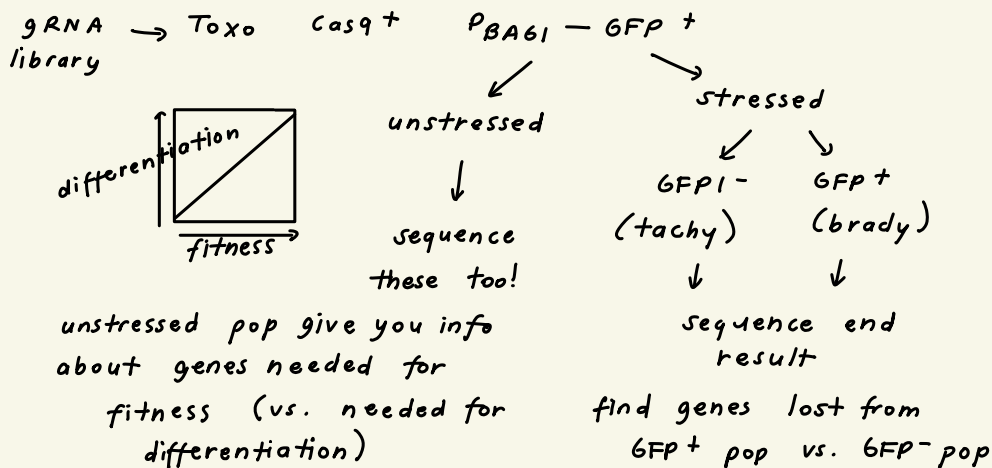
- major virulence behavior
- alteration of host metabolism
 - toxo secretes proteins that insert PV membrane + bind to mitochondrial proteins + bring close to PV
 - probably hijack mitochondria
- immune invasion
 - ROP16 kinase
 - highly variable across toxo strains
 - inc. phosphorylation of STAT6 TF
 - inc. host arginase $\rightarrow$ $\downarrow$ arginine $\rightarrow$ $\downarrow$ host cell NO production
 - disable innate immune path
 - losing arginine
 - amino acid starvation trigger tachy $\rightarrow$ bradyzoite transition
- immune activation
 - make sure the host survives long enough
 - $\uparrow$ IFN γ activation (immune cytokine)
 - more resistant to other pathogens
 - bacterial + viral pathogens
 - more long lived + likely to be eaten by cats

PAMPs $\rightarrow$ PRRs $\rightarrow$ MyD88 $\rightarrow$ IL-12 $\rightarrow$ IFN- γ $\rightarrow$ macrophage activation $\rightarrow$ toxoplasma control

D. tachy $\rightarrow$ bradyzoites

- ① Reliable way to tell them apart
 - GFP reporter behind BAg1 (on brdy)
- ② make perturbations
 - cas9 express constitutively + library of gRNAs

Approach:



look for enrichment on one side of diagonal

- hypothesis generator

- how to show necessary + sufficient?

- KO (deletion) for necessity
 - look for ability to form cysts
- bradyzoites not naturally forming but force it to happen w/ gene of interest

- what is upstream of this gene?

9.26 Rec 1 -

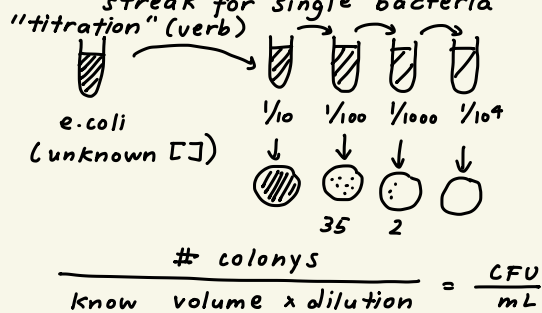
Methods:

- How did they assay for phage activity?
 - flow cytometry w/ GFP virus
 - take out a gene
 - RT-qPCR *how much viral DNA in cells?*
 - quantify viral DNA in host
 - RT b/c its an RNA virus
 - plaque assay *how many viral units?*
 - quantifying # of plaque-forming units (viable virus)
 - fluorescent antibodies

CFU: colony forming unit

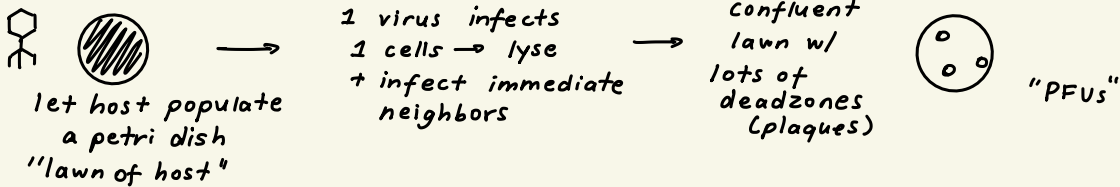
- foundational unit of bacterial microbiology

- streak for single bacteria



plaque assays: viruses are obligate parasites

- can't just put them on nutrient-rich media



- can now titrate successively to get exact []
- gold standard for infectious agent quantification

① compare PFUs across test condition

- change in the host

$$\frac{\text{\# PFUs in test condition}}{\text{\# PFUs in CHL}} = \% \text{ survival}$$

LRP knockout:

fewer plaques formed w/ partial infection knock-out

multiplicity of infection:

- desired ratio of virus to host
- average # virus / host
- MOI 1 = 1 virus / host
- $\neq$ exact number of viruses per host

- low MOI: likely a single hit
- high MOI: want every host infected

- can potentially compare qPCR instead of plaque assay

Paper Review:

How to identify host Receptor?

① Targeted mutagenesis screen

→ which genes are necessary for infection?

· no phenotype w/o the gene

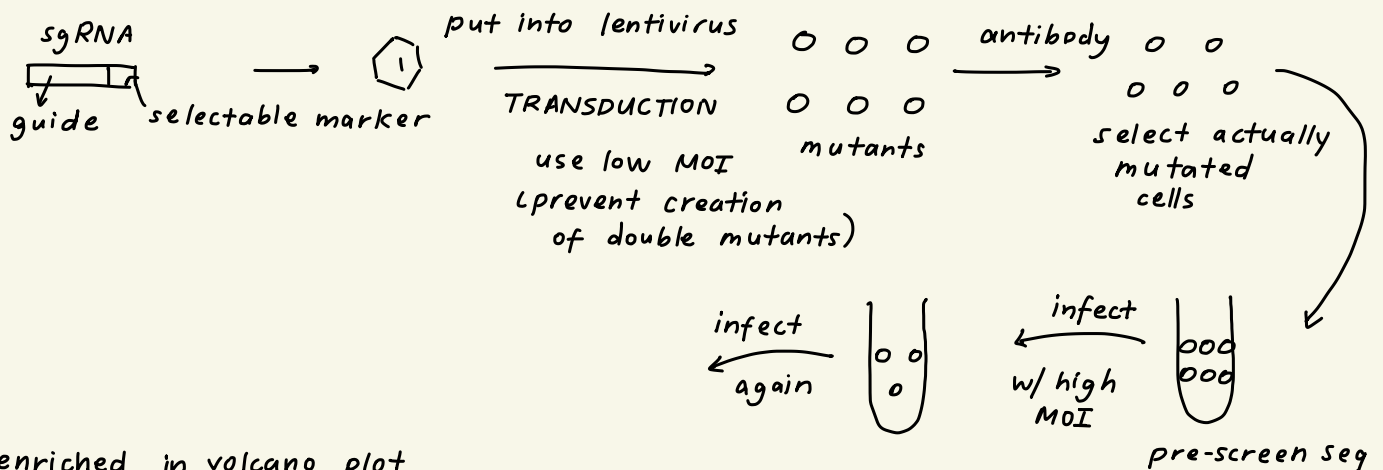
→ knock out every gene and see what isn't infected

② untargeted mutagenesis screen

→ overexpression library in a non-permissive host

· which factors are sufficient for infection?

→ with this gene alone, can it perform the phenotype?



- enriched in volcano plot

· 'see more of the gene (enriched)

· might alternatively be looking for depletion

- nonprotein interaction might not be present

Fig 2:

- looking for viral infection w/ GFP

dosage effect is quite powerful

- what's the point of a hypermorph?

· powerful correlation

7.26 Rec 5 -

exam review

T follicular helper cells:

B cell: $T_H F$

- ① co-stimulation signal
- ② somatic hypermutation
- ③ class switching (Ig type changes)

↓ ↘
memory cells plasma cells

dendritic cells

display antigen

B cells compete

to see who can

bind the strongest

↓

SOMATIC

HYPERMUTATION

(induce mutation in the variable region)

susceptibility vs. permissivity

Dap E $\rightarrow$ Dap F $\rightarrow$ Lys A $\rightarrow$ Lys - L

- necessary: if knock out, the process can't happen
- sufficiency: $\uparrow$ expression is enough the phenotype
- dap F is necessary for lysine but not sufficient!

evolutionary biology:

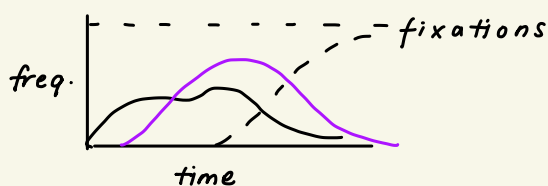
synonymous vs. nonsynonymous mutations

- hit a redundant codon

- rate of mutation: $\frac{dn}{ds}$

- clonal inference:

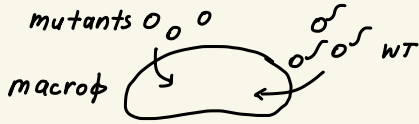
- 2 mutations that are independently beneficial for fitness
- these 2 mutations are competing $\rightarrow$ they eventually combine



- legionella pneumophila is an amoeba pathogen
- can also infect human macrophages

paper review:

- compete original bug against the mutated bug
- more mutations = better and better competitiveness
- lineages experience convergent evolution?
- validate which genes are actually involved
- why is the deletion of the flagella gene most competitive?



- then apply antibiotic
 - what doesn't get in gets wiped out
 - lyse & plate to see competitive ratio
- point mutation doing better subsequently
 - in the cell, this isn't actually true!

strain B:

- minimal unit that is sufficient?

WT = not auxotroph

w/ frameshifts that affect lysine prod.,

macrophage vs. amoeba fitness

- specialization occurring
- lysine metabolism necessary in amoeba

why do you need flagella as a legionella?

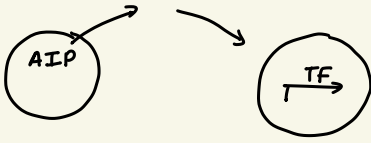
- water-borne
- it's not an obligate parasite so it needs the versatility offered by flagella

7.26 Review Session 2:

- Gram $\oplus$ vs. Gram $\ominus$

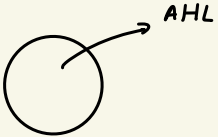
Quorum Sensing:

- Gram $\oplus$



- autoinducing proteins (AIP)
- no contact required

- Gram $\ominus$



things that enable bacteria to use humans as a niche
Virulence Factors:

- Toxin vs. Effector

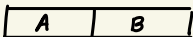
- both are bacterial proteins
 - independent of the host, secreted
 - active in the supernatant
 - directly damage/disrupt things
 - secreted into host cells to modulate host processes
- } TOXINS

- Koch's Molecular Postulates

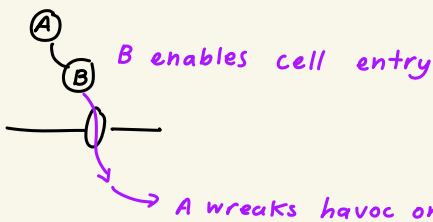
- ① gene is only present in pathogenic species
 - cholera gene phage
 - mobile elements in the pangenome
- ② loss of gene disrupt disease
- ③ genetic complementation: reintroduce the gene of interest
 - disease should return

- 3 major virulence factor classes:

- A/B toxins:
 - diphtheria + cholera toxins

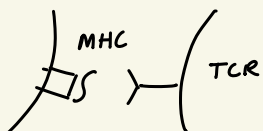


payload binding + entry



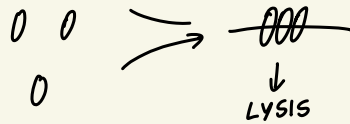
- superantigens

MHC I = CD8+ (all cells)
MHC II = CD4+ (in APCs)



- force interxn $\rightarrow$ proliferation cascade $\rightarrow$ immune rxn

- pore-forming toxins monomers

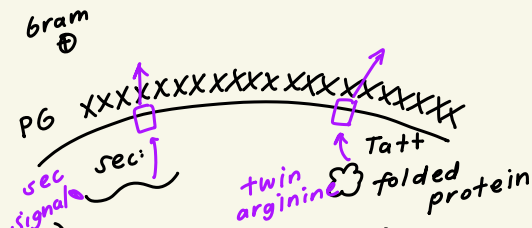


- for listeria, this enables endosome escape

Why lyse things?

- Nutrients (hemolysins to get Fe)
- Kill/disrupt immune cells
- Disrupt Barriers (Listeria via LLO)

Secretion Systems: TRANSPORT



- 1) 1 membrane crossing

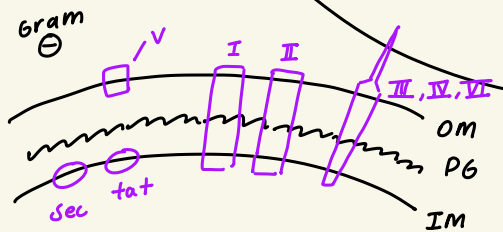
· sec/tat pathway:

- crossing the OM - T5 (sec dependent)

- IM/OM - T1, T2SS

- IM/OM/Host Membrane - T3, T4, T6SS

· use to secrete effectors into the host



☆ know stain type of common bacteria for the exam

TB:

- intracellular
- neither $\oplus$ or $\ominus$
- persistent
- respiratory disease

Intracellular/Extracellular Pathogens:

- Intracellular

- Listeria (gram $\oplus$)

- needs to enter the host
- receptor-mediated endocytosis
- escape phagosome to avoid getting digested
 - use pore-forming toxins
- move by hijacking actin
 - mimicking N-WASP (intracellular actin polymerization)

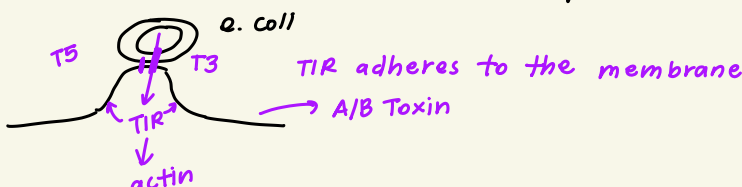
- salmonella

- flagellated for movement
- enter the cell to release VFs
- use type 3 secretion system for invasion + infection

- Extracellular

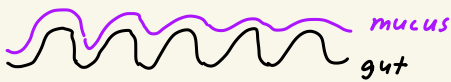
- E. coli

- most studied
- enteropathogenic E. coli
- needs to adhere to the outside of cells



challenges faced by Extracellular Pathogens:

① Mucus:



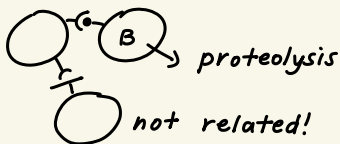
- detect mucus w/ metabolite presence
- fucose → T3SS to prevent colonization of mucus covered area
 - ↳ generated by commensals in the presence of mucus

② Limited Nutrients:

- iron
 - hemolysins
 - siderophores: high Fe affinity molecules
- "straw" like structures to "suck up" nutrients

③ microbial barriers

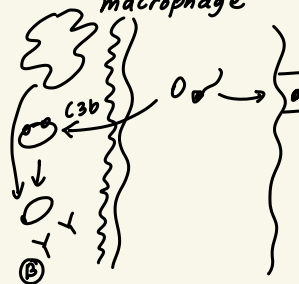
- how do you "win" against commensals?
 - bacteriocin
 - metabolites
] from commensals
- how to sense other cells?
 - quorum sensing
 - epinephrine
 - downstream activation of a pathogenicity island
 - contact dependent inhibition (CDi)



Immunity:

macrophage

within the gut



① Barriers

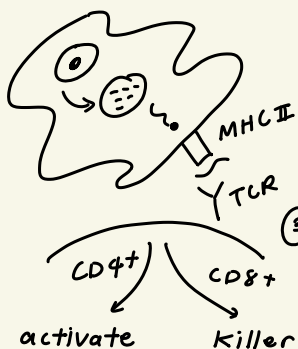
- mucus (physical)
- epithelial cell
- native microbiota
 - commensals, nutrient limitation

② Innate Immunity (immediately curb)

- complement: protein cascades
 - ① lysis w/ the MAC
 - ② opsonization
 - ③ inflammation
- APCs: phagocytes, NKs, Mφ
 - C3b in particular
 - ↳ MHCII interact w/ helper T cells
- bacteriocins: Antimicrobial peptides
- PAMPs/DAMPs
 - broad triggers (PRRs)
 - look for disease phenotypes
 - inflammasomes → pyroptosis

③ Adaptive Immunity

- Ab to neutralize/opsonize
- CD4+ helper T cells
 - ↳ activate immune system



evasion tactics

- cleave c3b

- downregulate MHC II

- lyse APCs

- ↓ likelihood of being lysed

- protease against antimicrobial peptides

- change surface charge

- downregulate PAMPs as needed